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The pseudo-arc from Bing's chains: uniqueness and homogeneity, following the machine-checked proofs

Scope and status. This document is an account of Theorems 1–6 and 10–13 of R. H. Bing, *A homogeneous indecomposable plane continuum*, Duke Math. J. 15 (1948), 729–742, in the form in which they have been machine-checked in the Lean 4 repository `HPC` (Mathlib; toolchain `leanprover/lean4:v4.28.0`). Every numbered Definition, Lemma, Proposition and Theorem below has a Lean counterpart, listed in Appendix A. The reported axiom profile of every theorem is `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`, no `native_decide` and no added axiom; the audits are Aristotle's `#print axioms` files listed in Appendix B and have not yet been reproduced independently. Until they are, "machine-checked" in this document means "reported as compiled with that audit".

What is and is not established. The two results that the classification of homogeneous plane continua takes from Bing's paper are Theorem 12 (any two continua admitting a *description*, in any compact metric spaces, are homeomorphic, with the two end points corresponding) and Theorem 13 (a continuum admitting a description is homogeneous). Both are proved here in full generality. The sentence "the pseudo-arc is a homogeneous plane continuum" needs one more fact: that a description *exists* in the plane. Bing's 1948 paper does not prove it; its "Description of M" paragraph posits the chains and points to a figure. The planar construction is Theorem 4 of Bing's *Snake-like continua* (Duke Math. J. 18 (1951), 653–663), and it has been machine-checked in the same repository (batch 84: `Plane.exists_description_plane`, and `Plane.pseudoArc_exists`, which asserts a nondegenerate, hereditarily indecomposable, homogeneous continuum in $\mathbb{R} \times \mathbb{R}$, from `theorem10` and `theorem13` of this document). That construction is not expounded in this document; Appendix B records its audit. Bing's *Concerning hereditarily indecomposable continua* (Pacific J. Math. 1 (1951), 43–51), Theorem 1 — any two nondegenerate chainable hereditarily indecomposable continua are homeomorphic, with points of different composants corresponding — has also been machine-checked (batch 88: `Plane.theorem1_HIC`, `Plane.pseudoArc_unique`), so the plane continuum of batch 84 is, up to homeomorphism, *the* pseudo-arc, and Theorem 13 of this document applies to every pseudo-arc.

Not treated are Theorems 7, 8 and 9, which serve only the stronger homogeneity statements, and

Theorems 14 and 15 themselves; Theorems 7–9 have since been machine-checked (batch 82; Bing's Theorem 8 needed correction and its literal form is refuted), Theorems 14–15 are in progress, and none of them is needed for Theorems 12 and 13. Bing's Theorem 3 carries a fourth conclusion (the end-link clause) which is not part of the machine-checked statement; it is not used by anything here.

Conventions. X is a metric space; where compactness is used it is stated. Indices of chains start at 0: Bing's link d_i is link $i-1$. In the formal development the fine chain is called E and the coarse chain D, the reverse of Bing's letters; this document uses the formal convention throughout and says so where Bing is quoted. A remark marked *Note* is commentary, not part of the argument.

Plan. Part I collects the definitions and Bing's Theorems 1 and 2. Part II is the proof of Theorem 3 (the fold), reproduced from the earlier stand-alone exposition with its numbering prefixed by "II.". Part III proves Theorem 4 from the fold, in the two forms Theorem 13 needs. Part IV proves Theorem 10. Part V proves Theorems 5 and 6 from one combinatorial statement. Part VI proves Theorems 11 and 12. Part VII proves Theorem 13, including the topological facts Bing's proof takes for granted, and ends with the dependency summary (§33). Appendix A maps every numbered item to the Lean development; Appendix B is the verification record. Where a proof below uses a hypothesis Bing does not state, or where a statement of Bing's is false as written and has been corrected, the paragraph says so.

Part I. Chains, crookedness, descriptions

1. Chains

Definition 1.1 (chain). A *chain* D of n links in X is a family of open sets $D_0, \dots, D_{n-1}$ such that $D_i \cap D_j \neq \emptyset$ if and only if $|i-j| \leq 1$. Taking $i=j$, every link is nonempty. Links are not assumed connected.

The *union* of D is $D^* = \bigcup_i D_i$. For $a \leq b$ the *subchain* $D(a,b)$ has links $D_a, \dots, D_b$; it is a chain of $b-a+1$ links. The *reverse* of D has links $D_{n-1}, \dots, D_0$.

Definition 1.2 (chain from P to Q). D is a *chain from P to Q* if $n \geq 1$, $P \in D_0$, $Q \in D_{n-1}$, P lies in no other link and Q lies in no other link.

Definition 1.3 (refinement, consolidation). A chain E *refines* D if every link of E lies in some link of D. F is a *consolidation* of E if E refines F and every link of F is a union of links of E.

Definition 1.4 (block consolidation). A *block map* on $\{0, \dots, n-1\}$ is a surjection β onto $\{0, \dots, p-1\}$ with $|\beta(t+1) - \beta(t)| \leq 1$ for all t. Its fibres are intervals of consecutive indices, the *blocks*. The *block consolidation* E/β is the chain of p links whose link u is $\bigcup_{\beta(t)=u} E_t$.

Lemma 1.5. E/β is a chain, and it is a consolidation of E.

Proof. Openness: a union of open sets. Adjacency: if links u, u' of E/β meet, some E_t with $\beta(t) = u$ meets some $E_{t'}$ with $\beta(t') = u'$, so $|t-t'| \leq 1$ and hence $|u-u'| \leq 1$. Conversely if $u' = u+1$, pick t_0, t_1 with $\beta(t_0) = u$, $\beta(t_1) = u+1$; walking from t_0 to t_1, β changes by at most one per step, so at the first index where it reaches $u+1$ the predecessor has value u; those two adjacent links of E meet, and their intersection lies in link u and link $u+1$ of E/β. The consolidation clauses are immediate. □

Note. Bing's consolidations (his D6) allow a link of F to be the union of an arbitrary subcollection of links of E. The formal development uses block consolidations throughout; every consolidation it produces is one, and every consolidation Theorems 12 and 13 consume is one. Where Bing's own constructions produce non-block consolidations (Theorem 7), this document does not follow them.

Definition 1.6 (pattern). E (with n links) *follows the pattern* $x : \{0, \dots, n-1\} \to \{0, \dots, m-1\}$ *in* D (with m links) if $E_t \subseteq D_{x(t)}$ for every t. A pattern in the sense of Theorem 6 is a map x with $x(0) = 0$, $x(n-1) = m-1$ and $|x(t+1) - x(t)| \le 1$.

Lemma 1.7 (patterns are Lipschitz). If E follows x in D then $|x(t+1) - x(t)| \le 1$ for all t: adjacent links of E meet, so the links of D containing them meet.

2. Crookedness

Definition 2.1 (between). For indices i, r, s, j of a chain, Between(i, r, s, j) means $i < r < s < j$ or $i > r > s > j$: r and s lie strictly between i and j, with r nearer i.

Definition 2.2 (crooked, Bing 1948). E is *crooked in* D if E refines D and for all links D_h, D_k of D with $k > h+2$, and all links E_i, E_j of E with $E_i \cap D_h \neq \emptyset$ and $E_j \cap D_k \neq \emptyset$, there are indices r, s with Between(i, r, s, j), $E_r \subseteq D_{k-1}$ and $E_s \subseteq D_{h+1}$.

In words: going from a link that meets D_h to a link that meets D_k, the chain E first enters the link of D next to D_k and afterwards the link of D next to D_h. (Bing's formulation "$E(i,j)$ is the sum of three chains $E(i,r)$, $E(r,s)$, $E(s,j)$ with $(s-r)(j-i) > 0$" is the same condition.)

Theorem 2.3 (Bing, Theorem 1). If D is a consolidation of E and F is crooked in E, then F is crooked in D.

Proof. F refines E, which refines D. Let $k > h+2$, $F_i \cap D_h \neq \emptyset$, $F_j \cap D_k \neq \emptyset$; say $i < j$. Replacing i by the last index before j whose link meets D_h, and then j by the first index after i whose link meets D_k, we may assume no interior link of $F(i,j)$ meets D_h or D_k. For each link F_t let $\delta(t)$ be an index of a link of D containing it; δ moves by at most one per step (Lemma 1.7). A link F_t with $\delta(t) = h$ meets D_h, so δ never takes the value h or k on the interior of $F(i,j)$, and since it starts within one of h and ends within one of k it stays in $[h+1, k-1]$ there.

Since D_h and D_k are unions of links of E, there are links $E_a \subseteq D_h$ meeting F_i and $E_b \subseteq D_k$ meeting F_j. Each link of $F(i,j)$ lies in a link of E, and the index of that link moves by at most one per step, so every index between a and b is the index of a link of E containing some link of $F(i,j)$; every such link of E lies in a link of D meeting a link of D with index in $[h+1, k-1]$ (or is E_a, E_b), hence lies in $D(h,k)$. The link $E_{a'}$ of $E(a,b)$ adjacent to E_a contains an interior link of $F(i,j)$ (the first link of $F(i,j)$ not inside E_a), which misses D_h; so $E_{a'} \not\subseteq D_h$, and since $E_{a'}$ meets $E_a \subseteq D_h$ and lies in $D(h,k)$, $E_{a'} \subseteq D_{h+1}$. Likewise the link $E_{b'}$ adjacent to E_b lies in D_{k-1}.

Now F is crooked in E: between F_i (meeting E_a) and F_j (meeting E_b), with $|a-b| > 2$ (else D_h and D_k would be joined by at most three consecutive links of E, whose middle link would lie in a link of D meeting both), there are r, s between i and j with $F_r \subseteq E_{b'}$ and $F_s \subseteq E_{a'}$. Then $F_r \subseteq D_{k-1}$ and $F_s \subseteq D_{h+1}$. □

Note. Bing remarks that Theorem 1 fails if "D is a consolidation of E" is weakened to "D contains E". The formal statement is for consolidations.

Theorem 2.4 (Bing, Theorem 2). If F refines E and E is crooked in D, then F is crooked in D.

Proof. F refines D by transitivity. Given $F_i \cap D_h \neq \emptyset$ and $F_j \cap D_k \neq \emptyset$ with $k > h+2$, let $E_a \supseteq F_i$ and $E_b \supseteq F_j$; then E_a meets D_h and E_b meets D_k, so crookedness of E in D gives $E_u \subseteq D_{k-1}$ and $E_v \subseteq D_{h+1}$ between a and b. Since F refines E and is a chain, $F(i,j)$ passes through E_u and E_v in order: the index of the link of E containing F_t moves by at most one per step (Lemma 1.7), so it takes every value between a and b, first u then v. The links $F_r \subseteq E_u$ and $F_s \subseteq E_v$ so found are the witnesses. □

Corollary 2.5. If E is crooked in D and E is a consolidation of F, then F is crooked in D. (Bing's corollary to Theorem 2: F refines E.)

Lemma 2.6 (subchains and reversal). If E is crooked in D then every subchain of E is crooked in D; E reversed is crooked in D; E is crooked in D reversed. If moreover every link of E lies in the subchain $D(a,b)$, then E is crooked in $D(a,b)$.

Proof. The witnesses of Definition 2.2 lie strictly between the two given links, hence in the subchain; reversal reverses "between" and exchanges the roles of h and k. □

3. Descriptions

Definition 3.1 (description). A *description* Δ of a continuum from P to Q in X consists of a sequence of chains $D^{(0)}, D^{(1)}, \ldots$ (Bing's $D_1, D_2, \ldots$) with n_i links, and a sequence $\varepsilon_i > 0$ with $\varepsilon_i \to 0$, such that for every i:

1. $D^{(i)}$ is a chain from P to Q;
2. $D^{(i+1)}$ is crooked in $D^{(i)}$;
3. the closure of every link of $D^{(i+1)}$ lies in some link of $D^{(i)}$;
4. every link of $D^{(i)}$ is bounded with diameter at most ε_i.

The *continuum described* is $M = \bigcap_i (D^{(i)})^*$.

Note. Bing's condition "the closure of each link of D_{i+1} is a compact subset of a link of D_i" is (3) together with compactness of X, assumed where needed. Bing's mesh bound $1/i$ is any $\varepsilon_i \to 0$. Bing's D_1 is $D^{(0)}$.

Lemma 3.2. $P, Q \in M$; M is closed, and compact when X is; $M \subseteq (D^{(i)})^*$ for every i; $M = \bigcap_i \overline{(D^{(i)})^*}$; every link of every $D^{(i)}$ meets M (when X is compact).

Proof. $P \in D_0^{(i)}$ for all i. For closedness: $(D^{(i+1)})^* \subseteq \bigcup_t \overline{D_t^{(i+1)}} \subseteq (D^{(i)})^*$ by (3), so the closures of the unions are nested and $\bigcap_i (D^{(i)})^* = \bigcap_i \overline{(D^{(i)})^*}$, an intersection of closed sets. That each link of $D^{(i)}$ meets M: a link $D_s^{(i)}$ contains a link of $D^{(i+1)}$ (the chain $D^{(i+1)}$ from P to Q passes through every link of $D^{(i)}$, by Lemma 1.7 applied to the refinement indices, which go from 0 to $n_i - 1$), which contains the closure of a link of $D^{(i+2)}$, and so on; the nested nonempty compact closures have a common point, which lies in M. □

Definition 3.3 (shift). For $i \geq 0$, the *tail* $\Delta_{\geq i}$ is the description with chains $D^{(i)}, D^{(i+1)}, \ldots$. It describes the same M.

Lemma 3.4 (crookedness down the description). For $j > i$, $D^{(j)}$ is crooked in $D^{(i)}$.

Proof. $D^{(j)}$ refines $D^{(j-1)}$, which refines ... $D^{(i+1)}$, which is crooked in $D^{(i)}$; apply Theorem 2.4. □

Lemma 3.5 (thick readings). Let X be compact and $w \geq 0$. For all large m: for every position p of $D^{(m)}$ there is a link s of $D^{(i)}$ such that every link of $D^{(m)}$ within w positions of p lies in $D^{(i)}_s$.

Proof. The links of $D^{(i)}$ cover the compact set $\overline{(D^{(i+1)})^*}$ (by (3)); let δ be a Lebesgue number. For $m > i$ with $\varepsilon_m(w+2) < \delta$, the links of $D^{(m)}$ within w positions of p lie in the ball of radius $(w+1)\varepsilon_m < \delta$ about a point of $D^{(m)}_p$ (points of links at index distance k are within $(k+1)\varepsilon_m$), which lies in a single link of $D^{(i)}$. □

Note. Lemma 3.5 is the form in which compactness enters Theorems 5, 6 and 13. Bing's "five-linked subchains" in the proof of Theorem 5 are this lemma with $w = 4$; the proofs below need w as large as the pattern.

Part II. Theorem 3: the fold, by the greedy jump zigzag

This Part reproduces, with its numbering prefixed by "II.", the self-contained exposition of the machine-checked proof of Bing's Theorem 3. Its Definitions II.0.1–II.0.2 restate Definitions 2.2 and 1.4 of Part I in the notation used there; §II.1 constructs the *faithful level model* (ρ, σ) of a crooked refinement, and §II.2 the *folds* that Parts III and VII consume. Theorem II.0.3 is what Part III builds on; the rest of the document cites items of this Part as "Lemma II.1.1", "Proposition II.6.5", and so on.

II.0 Statement

Throughout, X is a metric space. A **chain** D of k links is a finite sequence $D_0, \dots, D_{k-1}$ of subsets of X with $D_h \cap D_{h'} \neq \emptyset$ exactly when $|h - h'| \leq 1$. A chain E of n links **refines** D if every link of E lies in some link of D.

Definition II.0.1 (crooked, Bing 1948). E is *crooked in* D if E refines D and, whenever $h + 2 < k$ and links E_i, E_j meet D_h, D_k respectively, there are indices r, s between i and j (in the order i, r, s, j or j, s, r, i) with $E_r \subseteq D_{k-1}$ and $E_s \subseteq D_{h+1}$.

Definition II.0.2 (block consolidation). A *block map* on $\{0, \dots, n-1\}$ is a surjection $\beta' : \{0, \dots, n-1\} \to \{0, \dots, q-1\}$ whose fibres are intervals of consecutive indices and which is monotone in the sense that $|\beta'(t+1) - \beta'(t)| \leq 1$ for all t. The *consolidation* E/β' is the chain of q links whose u-th link is $\bigcup_{\beta'(t)=u} E_t$.

Theorem II.0.3 (Bing 1948, Theorem 3; block reading, three-link form). Let E be crooked in D, with $n \geq 1$ links, and let i be an index of E. Then there is a block map β' with $\beta'(i) = 0$ such that every link of E/β' lies in the union of three consecutive links of D.

The proof has two parts. §II.1 replaces chains by integer sequences: it is a reduction, and it uses nothing about crookedness beyond reading Definition II.0.1 off the sequences. §§II.2–II.6 is the theorem about sequences.

Sections II.1–II.2 reduce Theorem II.0.3 to a statement about integer sequences.

II.1 The level sequence of a refinement

Fix E crooked in D. For each index t of E put

$$\rho(t) = \min\{h : E_t \subseteq D_h\}, \qquad \sigma(t) = \max\{h : E_t \subseteq D_h\}.$$

Both are defined since E refines D.

Lemma II.1.1 (the faithful model). The pair (ρ, σ) satisfies:

1. $\rho(t) \le \sigma(t) \le \rho(t) + 1$;
2. $|\rho(t+1) - \rho(t)| \le 1$;
3. (*crookedness on levels*) for indices $s < t$ and integers h, k with $|h - k| \ge 3$ such that some u with $|u - s| \le 1$ has $\rho(u) \le h \le \sigma(u)$ and some u' with $|u' - t| \le 1$ has $\rho(u') \le k \le \sigma(u')$, there are $s < r < r' < t$ with
   - if $h < k$: $\rho(r) \le k - 1 \le \sigma(r)$ and $\rho(r') \le h + 1 \le \sigma(r')$;
   - if $k < h$: $\rho(r) \le k + 1 \le \sigma(r)$ and $\rho(r') \le h - 1 \le \sigma(r')$.

Proof. (1) A link of E lying in two links of D lies in their intersection, so those links are adjacent. (2) E_t and E_{t+1} meet, so $D_{\rho(t)}$ and $D_{\rho(t+1)}$ meet. (3) is Definition II.0.1 read off ρ, σ: a link carrying level h within one index of s meets D_h, and $E_r \subseteq D_{k-1}$ says $\rho(r) \le k - 1 \le \sigma(r)$. □

Property (1) is what "level" means: a link's levels form an interval $[\rho(t), \sigma(t)]$ of length at most one. Only (1)–(3) are used below; the geometric adjacency of consecutive intervals, which the repository also records, plays no part in this proof.

II.2 Folds, and the reduction

Definition II.2.1 (arms). For a sequence $f : \{0, \dots, n-1\} \to \mathbb{Z}$ and a basepoint i, the *forward arm* is $j \mapsto f(i + j)$ for $0 \le j < n - i$ and the *backward arm* is $j \mapsto f(i - j)$ for $0 \le j \le i$. Both start at $f(i)$.

Definition II.2.2 (fold). Let $g : \{0, \dots, m-1\} \to \mathbb{Z}$ be a sequence with $|g(j+1) - g(j)| \le 1$. A *walk* is a pair (ℓ, c) with $\ell \ge 1$ and $c : \mathbb{N} \to \mathbb{Z}$. The sequence g *folds onto* (ℓ, c) if there is an *itinerary* $\beta : \{0, \dots, m-1\} \to \{0, \dots, \ell - 1\}$ with

$$\beta(0) = 0, \qquad |\beta(j+1) - \beta(j)| \le 1, \qquad |g(j) - c(\beta(j))| \le 1 \text{ for all } j.$$

Note. No condition is placed on c: consecutive values of the walk may differ by any amount. The itinerary is what is Lipschitz. §§II.3–II.6 use this freedom.

Proposition II.2.3 (the reduction). Let E be crooked in D and let (ρ, σ) be its level sequence. Let i be an index and suppose there is one walk (ℓ, c) onto which both arms of ρ at i fold. Then the conclusion of Theorem II.0.3 holds at i.

Proof. Let β_G, β_T be itineraries for the forward and backward arms. Define $\beta'(t) = \beta_G(t - i)$ for $t \ge i$ and $\beta'(t) = \beta_T(i - t)$ for $t \le i$; the two agree at $t = i$, where both are 0. Then $|\beta'(t+1) - \beta'(t)| \le 1$, $\beta'(i) = 0$, and β' takes every value between 0 and its maximum, so β' is a block map onto $\{0, \dots, q-1\}$ with $q = 1 + \max \beta'$. For a block u, every t in the fibre has $|\rho(t) - c(u)| \le 1$, so $E_t \subseteq D_{\rho(t)}$ lies in the union of the three links of D with indices in $\{c(u) - 1, c(u), c(u) + 1\}$ (clipped to the range of D). □

Proposition II.2.4 (two-level bands). If $\max\rho - \min\rho \le 2$, the conclusion of Theorem II.0.3 holds at every i, with $q = 1$.

Proof. Every link of E lies in a link of D with index in $[\min\rho, \min\rho + 2]$. □

Theorem II.0.3 therefore follows from:

Theorem II.2.5 (S3). Let (ρ, σ) satisfy Lemma II.1.1 (1)–(3) on $\{0, \dots, n-1\}$, with $L \le \rho \le H$ and $H \ge L + 3$. Then for every basepoint i there is a walk onto which both arms of ρ at i fold.

Proof of Theorem II.0.3 from Theorem II.2.5. Form (ρ, σ) by §II.1; put $L = \min\rho$, $H = \max\rho$. If $H \le L + 2$ apply Proposition II.2.4. Otherwise apply Theorem II.2.5 and Proposition II.2.3. □

The rest of this document proves Theorem II.2.5.

Sections II.3–II.6 prove that statement (Theorem II.2.5).

II.3 Crookedness across the basepoint

Fix (ρ, σ) and i as in Theorem II.2.5. Re-index at the basepoint: $S(x) = \rho(i + x)$ for $-i \le x \le n - 1 - i$ (and $S(x) = 0$ elsewhere, a value never used). The backward arm is $j \mapsto S(-j)$ on $[0, M_-)$ with $M_- = i + 1$, the forward arm $j \mapsto S(j)$ on $[0, M_+)$ with $M_+ = n - i$; both are 1-Lipschitz, take values in $[L, H]$, and start at $p = S(0)$.

Definition II.3.1 (crookedness on levels). A function $S : \mathbb{Z} \to \mathbb{Z}$ is *crooked on* $[\mathrm{lo}, \mathrm{hi}]$ if for all $\mathrm{lo} \le s < t \le \mathrm{hi}$:

- if $S(s) + 3 \le S(t)$, there are $s < r < r' < t$ with $S(r) \in \{S(t) - 2, S(t) - 1\}$ and $S(r') \in \{S(s), S(s) + 1\}$;
- if $S(t) + 3 \le S(s)$, there are $s < r < r' < t$ with $S(r) \in \{S(t), S(t) + 1\}$ and $S(r') \in \{S(s) - 2, S(s) - 1\}$.

In words: between two times whose levels differ by three or more, the sequence first comes within one of the later level and then within one of the earlier one, in that order, and "within one" is on the side facing the other level.

Lemma II.3.2. S is crooked on $[-i, n - 1 - i]$.

Proof. Apply Lemma II.1.1 (3) with $h = S(s)$, $k = S(t)$ and the witnesses $u = s$, $u' = t$ themselves (each link carries its own ρ). If $h < k$ the clause gives r with $\rho(r) \le k - 1 \le \sigma(r) \le \rho(r) + 1$, so $\rho(r) \in \{k - 2, k - 1\}$, and r' with $\rho(r') \le h + 1 \le \rho(r') + 1$, so $\rho(r') \in \{h, h + 1\}$. The case $k < h$ is the same. □

Lemma II.3.3 (reversal). If S is crooked on $[\mathrm{lo}, \mathrm{hi}]$ then $x \mapsto S(-x)$ is crooked on $[-\mathrm{hi}, -\mathrm{lo}]$.

Proof. Reversing the order exchanges the roles of s and t and of r and r', and exchanges the two clauses. □

The proof of Theorem II.2.5 uses crookedness through exactly two configurations, which it shows to be impossible. In both, one arm is read as the *left* arm $j \mapsto S(-j)$ and the other as the *right* arm $j \mapsto S(j)$; by Lemma II.3.3 either arm may be taken as the left one.

Lemma II.3.4 (the up core). Let S be crooked on $[\mathrm{lo}, \mathrm{hi}]$, let $b \le a$, and let τ, d_r, d be times with $-d_r \ge \mathrm{lo}$, $d \le \mathrm{hi}$, such that

- the left arm is $\geq b$ at every time $< \tau$, is $\leq a-1$ at every time in $[\tau, d_r)$, and equals $b-2$ at d_r;
- the right arm is $\geq b$ at every time $< d$ and equals $a+2$ at d.

This is impossible.

Proof. The two times $-d_r < d$ have $S(-d_r) = b-2$ and $S(d) = a+2$, differing by at least 4. The first clause of Definition II.3.1 gives $-d_r < r < r' < d$ with $S(r) \in \{a, a+1\}$ and $S(r') \in \{b-2, b-1\}$. If $r' \geq 0$, it is a time of the right arm before d, where the arm is $\geq b$: contradiction. So $r' < 0$, and $-r'$ is a time of the left arm before d_r at level $< b$, hence $-r' \geq \tau$; then $-r$ lies in $(-r', d_r) \subseteq [\tau, d_r)$, where the left arm is $\leq a-1$, but $S(r) \geq a$. □

Lemma II.3.5 (the down core). Let S be crooked on $[\mathrm{lo}, \mathrm{hi}]$, let $b \leq a$, and let $\tau_b \leq \varsigma \leq \mathrm{rise}$ and $\tau_b' \leq d$ be times with $-\mathrm{rise} \geq \mathrm{lo}$, $d \leq \mathrm{hi}$, such that

- the left arm is $\geq b$ before τ_b, is $\leq a-2$ on $[\tau_b, \varsigma)$, is $\geq b+1$ on $[\varsigma, \mathrm{rise})$, and equals $a+2$ at rise;
- the right arm is $\geq b$ before τ_b', is $\leq a-1$ on $[\tau_b', d)$, and equals $b-2$ at d.

This is impossible.

Proof. Read crookedness between $-\mathrm{rise}$ and d (second clause, since $S(-\mathrm{rise}) = a+2 \geq S(d)+4$): there are $-\mathrm{rise} < r < r' < d$ with $S(r) \in \{b-2, b-1\}$ and $S(r') \in \{a, a+1\}$.

If $r \geq 0$, then r is a time of the right arm before d at level $< b$, so $r \geq \tau_b'$; then $r' \in (r, d) \subseteq [\tau_b', d)$, where the right arm is $\leq a-1$, but $S(r') \geq a$. Contradiction.

If $r < 0$, put $u = -r$, a time of the left arm before rise at level $\leq b-1$. Then $u \geq \tau_b$ (before τ_b the arm is $\geq b$) and $u < \varsigma$ (on $[\varsigma, \mathrm{rise})$ the arm is $\geq b+1$). Read crookedness a second time, between $-\mathrm{rise}$ and r (second clause again, $S(-\mathrm{rise}) = a+2 \geq S(r)+3$): there are $-\mathrm{rise} < q < q' < r$ with $S(q') \in \{a, a+1\}$. Both $-q$ and $-q'$ are left-arm times in (u, rise). The time $-q' > u \geq \tau_b$ has level $\geq a$, so it is not in $[\tau_b, \varsigma)$, hence $-q' \geq \varsigma$; then $-q \in (-q', \mathrm{rise}) \subseteq [\varsigma, \mathrm{rise})$, where the arm is $\geq b+1$, but $S(q) \leq S(r)+1 \leq b$. Contradiction. □

Note. Lemma II.3.4 reads crookedness once, across the basepoint. Lemma II.3.5 reads it twice: once across the basepoint and once inside the left arm. These two lemmas are the only use of crookedness in §§II.3–II.6.

II.4 Jump walks and the phase condition

Definition II.4.1 (jump walk). For integers a, b with $|b-a| \geq 3$, the *leg from a to b* has length $\lambda(a,b) = |b-a| - 2$ and consists of the values

$$a,\ a+3\varepsilon,\ a+4\varepsilon,\ \ldots,\ b-\varepsilon \qquad (\varepsilon = \mathrm{sign}(b-a)),$$

that is, the value at position u is a for $u = 0$ and $a + \varepsilon(2+u)$ for $1 \leq u < \lambda(a,b)$. For a start a_0 and turning levels $a_1, \ldots, a_K$ with $|a_{j+1} - a_j| \geq 3$, the *jump walk* $\mathrm{jwalk}(a_0; a_1, \ldots, a_K)$ is the concatenation of the legs $a_0 \to a_1$, $a_1 \to a_2$, ..., $a_{K-1} \to a_K$, followed by the single value a_K. Its length is $\ell = 1 + \sum_j \lambda(a_j, a_{j+1})$.

The leg from a to b omits the two values $a+\varepsilon$, $a+2\varepsilon$: it *jumps* from a to $a+3\varepsilon$. Consequently a sequence at level $a \pm 1$ is within one of the walk's value a at position 0, and every level strictly

between a and b other than $a \pm 1, a \pm 2$ is within one of exactly the values at positions $1, 2, \ldots$; the last position of the leg carries $b - \varepsilon$.

Definition II.4.2 (envelope, arrival). For a leg $a \to b$: $\mathrm{Env}(a, b)$ is the interval $[\min(a, b) - 1, \max(a, b) + 1]$; the *arrival level* is $\mathrm{arr}(a, b) = b - \varepsilon$, the level one short of b on the side of a.

Definition II.4.3 (the phase condition). Let g be a sequence on $[0, m)$. $\mathrm{Fit}(g; \tau, a; a_1, \ldots, a_K)$, for $K \geq 1$, holds if $|a_1 - a| \geq 3$ and either

- (*final leg*) $g(j) \in \mathrm{Env}(a, a_1)$ for all $\tau \leq j < m$; or
- (*passage*) there is $\tau' \in [\tau, m)$ with $g(\tau') = \mathrm{arr}(a, a_1)$, $g(j) \in \mathrm{Env}(a, a_1)$ for all $\tau \leq j \leq \tau'$, and $\mathrm{Fit}(g; \tau', a_1; a_2, \ldots, a_K)$.

Fit with an empty list of turning levels is false.

Lemma II.4.4 (fold from the phase condition). Let g be 1-Lipschitz on $[0, m)$, let $\mathrm{Fit}(g; \tau, a; a_1, \ldots, a_K)$ hold and $|g(\tau) - a| \leq 1$. Then there is $\beta : \mathbb{N} \to \mathbb{N}$ with $\beta(\tau) = 0$, $\beta(j) < \ell$ for $\tau \leq j < m$, $|\beta(j + 1) - \beta(j)| \leq 1$ for $\tau \leq j$, $j + 1 < m$, and $|g(j) - \mathrm{jwalk}(a; a_1, \ldots, a_K)(\beta(j))| \leq 1$ for $\tau \leq j < m$.

Proof. Induction on K. Define the *leg position* of a level x on the leg $a \to b$ as

$$\mathrm{pos}_{a,b}(x) = \max\left(0, \min(\varepsilon(x - a) - 1,\ \lambda(a, b))\right).$$

Three facts: $\mathrm{pos}_{a,b}(x) \leq \lambda(a, b)$; $|\mathrm{pos}_{a,b}(y) - \mathrm{pos}_{a,b}(x)| \leq 1$ when $|y - x| \leq 1$; and for $x \in \mathrm{Env}(a, b)$ the walk's value at position $\mathrm{pos}_{a,b}(x)$ is within one of x: position 0 carries the value a and is assigned exactly to $x \in \{a - \varepsilon, a, a + \varepsilon\}$; a position $1 \leq u < \lambda$ carries $a + \varepsilon(2 + u)$ and is assigned to $x = a + \varepsilon(1 + u)$; position λ carries the next turning level b and is assigned to $x \in \{b - \varepsilon, b, b + \varepsilon\}$. Also $\mathrm{pos}_{a,b}(g(\tau)) = 0$ since $|g(\tau) - a| \leq 1$, and $\mathrm{pos}_{a,b}(\mathrm{arr}(a, b)) = \lambda(a, b)$.

In the final-leg case put $\beta(j) = \mathrm{pos}_{a,a_1}(g(j))$. In the passage case obtain β' for $\mathrm{Fit}(g; \tau', a_1; a_2, \ldots)$ from the induction hypothesis (note $|g(\tau') - a_1| = 1$), and put $\beta(j) = \mathrm{pos}_{a,a_1}(g(j))$ for $j \leq \tau'$ and $\beta(j) = \lambda(a, a_1) + \beta'(j)$ for $j > \tau'$. At $j = \tau'$ the two formulas agree ($\lambda(a, a_1) + 0$), the Lipschitz condition across τ' follows from $\beta'(\tau') = 0$ and $|\beta'(\tau' + 1)| \leq 1$, and the value of the concatenated walk at position $\lambda(a, a_1) + v$ is the value of the tail walk at position v. □

Note. The jump is what makes Fit purely order-theoretic: an arm passes to the next leg exactly when it first reaches the arrival level, and no relation between the *speeds* of the two arms is needed. With a unit-step walk, an arm on the leg would have to be tracked level by level from a, and the position at which it may leave the leg would depend on where the other arm is.

II.5 One phase of the greedy zigzag

Two arms are in play, indexed by $e \in \{-, +\}$: F_e is 1-Lipschitz on $[0, M_e)$ with values in $[L, H]$. A *state* assigns to each arm either *finished* or *live at* τ_e, meaning the arm entered the current turning level at time τ_e.

Definition II.5.1 (up phase, not stuck). An *up phase* from bottom b with previous top a, where $b + 2 \leq a$, starts from a state in which every live arm has $|F_e(\tau_e) - b| \leq 1$. The phase is *not stuck* if no live arm e has a time $d_r \geq \tau_e$ with $F_e(d_r) = b - 2$ and $F_e \leq a - 1$ on $[\tau_e, d_r)$.

Lemma II.5.2 (first drop). If g is 1-Lipschitz, $g(\tau) \geq b-1$, and $g(j) < b-1$ for some $j \geq \tau$, then the least such j, call it d_r, has $d_r > \tau$, $g(d_r) = b-2$, and $g \geq b-1$ on $[\tau, d_r)$. □

Lemma II.5.3 (transfer). Let g be 1-Lipschitz on $[0, m)$, $t \geq b+3$, $g(\tau) \leq b+1$, and suppose every $j \geq \tau$ with $g(j) < b-1$ is preceded by some $j' \in [\tau, j)$ with $g(j') \geq t-1$. Then either $g \in [b-1, t-2]$ on all of $[\tau, m)$, or there is $\varsigma > \tau$ with $g(\varsigma) = t-1$ and $g \in [b-1, t-2]$ on $[\tau, \varsigma)$.

Proof. If g never reaches $t-1$ after τ, the hypothesis shows it never drops below $b-1$ either. Otherwise let ς be the first time $\geq \tau$ with $g \geq t-1$; before it, $g \leq t-2$, and $g \geq b-1$ because a drop would be preceded by a visit to $t-1$ before ς; $g(\varsigma) = t-1$ by the Lipschitz condition. □

Proposition II.5.4 (one up phase). Consider an up phase from b with previous top a that is not stuck. Then there is a turning level $t \geq a+1$ and a new state such that:

1. finished arms stay finished;
2. a live arm that becomes finished satisfies $F_e \in [b-1, t-2]$ on $[\tau_e, M_e)$;
3. a live arm that stays live, now at ς_e, satisfies $\tau_e < \varsigma_e < M_e$, $F_e(\varsigma_e) = t-1$, and $F_e \in [b-1, t-2]$ on $[\tau_e, \varsigma_e)$;
4. either every arm is now finished, or $t \leq H+1$ and there is a *determining arm* e_0, live before and after, with a time $d_r > \varsigma_{e_0}$ at which $F_{e_0}(d_r) = b-2$ and $F_{e_0} \in [b-1, t-1]$ on $[\tau_{e_0}, d_r)$.

Proof. Put big $= \max(a+1, H+2)$. To each arm assign a value v_e: if e is finished, or live but never below $b-1$ after τ_e, put $v_e =$ big; otherwise let d_r be the first drop (Lemma II.5.2), let μ be the maximum of F_e on $[\tau_e, d_r)$, and put $v_e = \mu+1$. Since the phase is not stuck, $\mu \geq a$, so $v_e \geq a+1$; and $\mu \leq H$, so $v_e \leq$ big. In every case: every drop of e below $b-1$ after τ_e is preceded by a visit to level $\geq v_e - 1$ (in the second case, the maximum $\mu = v_e - 1$ is attained before d_r, and every later drop is after d_r).

Put $t = \min(v_-, v_+)$; then $t \geq a+1$, and every drop of a live arm is preceded by a visit to level $\geq t-1$. Lemma II.5.3 (with $t \geq b+3$ since $a \geq b+2$) gives (2) or (3) for each live arm; declare the arm finished or live at ς_e accordingly. For (4): if $t <$ big, some arm e_0 has $v_{e_0} = t$ and is in the second case, with d_r, $\mu = t-1$; it does not fall under (2), since $F_{e_0}(d_r) = b-2 < b-1$; so it is live at ς_{e_0}, and $\varsigma_{e_0} < d_r$ because the level $t-1$ is attained on $[\tau_{e_0}, d_r)$; finally $t-1 = \mu \leq H$. If $t =$ big $\geq H+2$, no arm reaches $t-1 \geq H+1$, so every arm is finished. □

Proposition II.5.5 (one down phase). The reflection of Proposition II.5.4: a down phase from top a with previous bottom b, $b+2 \leq a$, starting from live arms with $|F_e(\tau_e) - a| \leq 1$, is *not stuck* if no live arm has a time $d_r \geq \tau_e$ with $F_e(d_r) = a+2$ and $F_e \geq b+1$ on $[\tau_e, d_r)$. If it is not stuck there is a turning level $t \leq b-1$ and a new state with (1); (2) with $[t+2, a+1]$; (3) with $F_e(\varsigma_e) = t+1$ and $[t+2, a+1]$; and (4) with $t \geq L-1$, $F_{e_0}(d_r) = a+2$ and $F_{e_0} \in [t+1, a+1]$ on $[\tau_{e_0}, d_r)$.

Proof. Apply Proposition II.5.4 to $-F_e$ with (a, b, H) replaced by $(-b, -a, -L)$. □

Nothing in §II.5 uses crookedness.

II.6 No phase is stuck, and the induction

Now F_- is the backward arm and F_+ the forward arm of S, so $F_e(j) = S(\mp j)$, and S is crooked on $[1-M_-, M_+-1]$ (Lemma II.3.2). For the lemmas of §II.3 applied with arm e as the left arm, use Lemma II.3.3.

Lemma II.6.1 (an up phase is never stuck). Consider an up phase from b with previous top a, in

a state satisfying: (*pre*) each live arm e is $\geq b$ at every time $< \tau_e$; and (*det*) some arm e_0 has a time d with $F_{e_0}(d) = a+2$, $F_{e_0} \geq b$ before d, and $\tau_{e_0} \leq d$ if e_0 is live. Then the phase is not stuck.

Proof. Suppose arm e is live at τ and has $d_r \geq \tau$ with $F_e(d_r) = b-2$ and $F_e \leq a-1$ on $[\tau, d_r)$. If $e = e_0$: $d \neq d_r$ (levels $a+2 \neq b-2$), and $d_r < d$ would put d_r before d at a level $< b$; so $d < d_r$, and $\tau \leq d$, so $F_e(d) \leq a-1$, contradicting $F_e(d) = a+2$. If $e \neq e_0$: take e as the left arm and e_0 as the right; then Lemma II.3.4 applies with τ, d_r, d (the left arm is $\geq b$ before τ by (*pre*)). □

Lemma II.6.2 (a down phase is never stuck). Consider a down phase from top a with previous bottom b, reached from an up phase from b in a state st (entry times τ_e) satisfying (*pre*) for b, and with the new entry times ς_e satisfying $F_e \leq a-2$ on $[\tau_e, \varsigma_e)$ for each arm live in both. Suppose (*det*): some arm e_0 live in both has a time $d > \varsigma_{e_0}$ with $F_{e_0}(d) = b-2$ and $F_{e_0} \leq a-1$ on $[\tau_{e_0}, d)$. Then the down phase is not stuck.

Proof. Suppose arm e, live at ς, has rise $\geq \varsigma$ with $F_e(\text{rise}) = a+2$ and $F_e \geq b+1$ on $[\varsigma, \text{rise})$. If $e = e_0$: rise $\neq d$, and $d <$ rise would put d in $[\varsigma, \text{rise})$ at level $b-2$; so rise $< d$, and rise $\geq \varsigma > \tau_{e_0}$, so $F_e(\text{rise}) \leq a-1$: contradiction. If $e \neq e_0$: take e as the left arm with $\tau_b = \tau_e$, ς, rise, and e_0 as the right arm with $\tau'_b = \tau_{e_0}$, d; Lemma II.3.5 applies. □

Lemma II.6.3 (the first phase). With $p = S(0)$, at least one of the following holds: (A) no arm has a time d_r with $F_e(d_r) = p-3$ and $F_e \leq p$ before d_r; (B) no arm has a time d with $F_e(d) = p+3$ and $F_e \geq p$ before d.

Proof. If both fail, with witnesses (e_1, d_1) for (A) and (e_2, d_2) for (B): if $e_1 = e_2$, whichever of d_1, d_2 comes first contradicts the bound on the other. Otherwise Lemma II.3.4 applies with e_1 as left arm, $\tau = 0$, $d_r = d_1$, $d = d_2$, $a = p+1$, $b = p-1$. □

Proposition II.6.4 (the main induction). Let $b+2 \leq a$, $a \leq H+1$, $b \geq L-1$, and let a state be given with: every live arm has $|F_e(\tau_e) - b| \leq 1$; (*pre*) for b; and the up phase from b with previous top a is not stuck. Then there is a list of turning levels $t_1, t_2, \ldots$ such that Fit$(F_e; \tau_e, b; t_1, t_2, \ldots)$ holds for every live arm e.

Proof. Strong induction on $N = (H+1-a) + (b-L+1) \geq 0$.

Run the up phase (Proposition II.5.4): turning level $t \geq a+1$, new state. For every live arm e the leg $b \to t$ has height ≥ 3; an arm finished by (2) satisfies the final-leg clause of Fit on this leg; an arm live at ς_e by (3) satisfies the passage clause with $\tau' = \varsigma_e$ (its level there is $t-1 = \text{arr}(b,t)$, and $[b-1, t-2] \cup \{t-1\} \subseteq \text{Env}(b,t)$), provided Fit$(F_e; \varsigma_e, t; \text{rest})$ holds for the remaining list. If every arm is finished, take the list $[t]$.

Otherwise (4) gives $t \leq H+1$ and a determining arm; (*pre*) and (3) give the hypotheses of Lemma II.6.2, so the down phase from t with previous bottom b is not stuck. Run it (Proposition II.5.5): turning level $t_2 \leq b-1$, $t_2 \geq L-1$, new state; the leg $t \to t_2$ is fitted by each arm as before. If every arm is now finished, take $[t, t_2]$.

Otherwise (4) of the down phase gives a determining arm e_1 with a time d_r at which $F_{e_1}(d_r) = t+2$, staying $\geq t_2+1$ on $[\varsigma_{e_1}, d_r)$. Establish (*pre*) for t_2: an arm live at ς'_e after the down phase is $\geq t_2$ before ς'_e — before τ_e it is $\geq b > t_2$, on $[\tau_e, \varsigma_e)$ it is $\geq b-1 \geq t_2$, and on $[\varsigma_e, \varsigma'_e)$ it is $\geq t_2+2$. The same bounds show $F_{e_1} \geq t_2$ before d_r. Lemma II.6.1 (with $a := t$, $b := t_2$, $d := d_r$) shows the next up phase is not stuck. The new parameters (t_2, t) satisfy $t_2+2 \leq t$, $t \leq H+1$, $t_2 \geq L-1$, and

$$(H+1-t)+(t_2-L+1)\le(H+1-a-1)+(b-L+1-1)<N,$$

so the induction hypothesis gives a list rest fitting every arm live at ς'_e from t_2; the list $[t,t_2]$ ++ rest fits every arm live at the start. □

Proposition II.6.5 (from the basepoint). There are T_0 with $|p-T_0|\le 1$ and a list $t_1,t_2,\dots$ such that Fit$(F_e;0,T_0;t_1,t_2,\dots)$ for both arms.

Proof. By Lemma II.6.3. In case (A), take both arms live at $\tau_e=0$ with $b=p-1$, $a=p+1$: then $|F_e(0)-b|=1$, (*pre*) is vacuous, and the up phase is not stuck because a drop to $b-2=p-3$ while staying $\le a-1=p$ is exactly what (A) excludes. Proposition II.6.4 gives the list; $T_0=p-1$. In case (B), take $a=p+1$, $b=p-1$ and run one down phase first (Proposition II.5.5, not stuck by (B)); if any arm survives, its determining arm and the bounds of the phase give (*pre*) for t_2 and, by Lemma II.6.1, a non-stuck up phase from t_2 with previous top $p+1$; Proposition II.6.4 finishes; $T_0=p+1$. □

Proof of Theorem II.2.5. Proposition II.6.5 and Lemma II.4.4 give, for each arm, an itinerary β_e with $\beta_e(0)=0$ onto the one jump walk jwalk$(T_0;t_1,t_2,\dots)$, of length ℓ; these are folds in the sense of Definition II.2.2. □

II.7 What the proof uses, and what it does not

- Of the faithful model, only Lemma II.1.1 (1)–(3): the interval structure of levels, the Lipschitz condition, and the crookedness clause read on the levels of the two endpoints. The geometric adjacency of consecutive intervals is not used.
- Of crookedness, exactly two readings (Lemmas II.3.4 and II.3.5), one of which reads the clause twice. No timing constant (the recurrence $1,2,3,6,13,30,\dots$ of Bing's Theorem 7) enters.
- Of the definition of a fold, the freedom to choose the walk c without a Lipschitz condition. Consecutive positions of the jump walk differ by 3 at each turn, so consecutive blocks of the consolidation in Theorem II.0.3 lie in adjacent but disjoint three-link windows of D at those positions.
- The span $H-L$ enters only through termination: the number of phases is at most the number of levels in the band, and the walk has at most one leg per phase.

Part III. Theorem 4: a consolidation from r to s

10. Statement

Theorem 10.1 (Bing, Theorem 4; block reading, three-link form). Let E be crooked in D, where D has $k\ge 3$ links, and let $r\le s$ be indices of E such that some link E_h with $r\le h\le s$ meets D_0 and some link $E_{h'}$ with $r\le h'\le s$ meets D_{k-1}. Then there is a block map β with $\beta(r)=0$, $\beta(s)=\max\beta$, and every block of E/β inside the union of three consecutive links of D.

Theorem 10.2 (two-link basepoints). Under the hypotheses of Theorem 10.1, with indices $r < r' = r+1 \le s' = s-1 < s$, the block map may be chosen with $\beta(r) = \beta(r') = 0$ and $\beta(s') = \beta(s) = \max \beta$, still with every block in three consecutive links of D.

Bing's statement (p. 733) is Theorem 10.1 with "two adjacent links" in place of three and "d_r contained in only the first link of F, d_s in only the last" in place of $\beta(r) = 0$, $\beta(s) = \max \beta$. Theorem 10.2 is not in Bing; Part VII explains why Theorem 13 needs it (Lemma 16.1).

11. Bing's proof, and what the block reading does instead

Bing derives Theorem 4 from Theorem 3 applied twice — at d_r to the tail $D(1,h)$, at d_s to the tail $D(k,n)$, where d_h meets e_1 and d_k meets e_m inside $D(r,s)$ — and then consolidates the links in between *by level* and splices the three pieces. The splice needs the block containing d_h to lie in the bottom window $e_1 + e_2$, which is the fourth conclusion of his Theorem 3 (the end-link clause).

The proof below follows the same plan with one change. It does not use the end-link clause. In its place, the fold of Part II is run on the tail with one *fake link* at level L appended after the link E_h meeting D_0; the fake link is admissible for crookedness precisely because E_h meets D_0; and the splice is cut not at the block of E_h but at the *time* U at which the fold reaches the fake link, where the walk's value is at most $L+1$ by construction.

Throughout §§12–14, $\rho(t)$ and $\sigma(t)$ are the least and greatest indices of links of D containing E_t (§II.1), so $0 \le \rho \le \sigma \le \rho + 1 \le k$, both are 1-Lipschitz, and $|\sigma(t+1) - \rho(t)| \le 1$ (adjacency of intervals, Lemma II.1.1 and the `adj` field of the faithful model).

12. The fold with a fake end

Lemma 12.1 (crookedness read at a link meeting D_0). Let E be crooked in D, $E_e \cap D_0 \neq \emptyset$, $t < e$ and $\rho(t) \ge 3$. Then there are $t < y < y' < e$ with $\rho(y) \le 1$ and $\rho(t) - 2 \le \rho(y') \le \rho(t) - 1$.

Proof. Definition 2.2 with $h = 0$ (met by E_e), $k = \rho(t)$ (met by E_t), $k > h + 2$: between e and t — in the order from t toward e — first a link inside D_{k-1}, then a link inside D_1; the first gives y' with $\rho(y') \in \{k-2, k-1\}$ (Lemma II.1.1(1)), the second y with $\rho(y) \le 1$. Reading Definition 2.2 with $i = e$, $j = t$ places r (inside D_{k-1}) nearer e and s (inside D_1) nearer t — so in the order from t: first D_1, then D_{k-1}. That is the order claimed. □

Definition 12.2 (the extended sequence). For a basepoint r and a later index a with $E_a \cap D_0 \neq \emptyset$, let S be the level sequence ρ re-indexed at r on $[-r, a-r]$, and let S' agree with S there and take the value 0 at $a - r + 1$: the fake link.

Lemma 12.3. S' is crooked (Definition II.3.1) on $[-r, a-r+1]$.

Proof. Pairs of times not involving the fake time are Lemma II.3.2. For a pair $(x, a-r+1)$ with $S(x) \ge 3$: Lemma 12.1 gives $y < y'$ strictly between with $S(y) \le 1$ and $S(y') \in \{S(x) - 2, S(x) - 1\}$, which is the second clause of Definition 3.1 with $S(t) = 0$ at the fake time (the clause asks $S(y) \in \{0,1\}$ and $S(y') \in \{S(x)-2, S(x)-1\}$). Pairs with $S(x) \le 2$ impose nothing. □

Proposition 12.4 (the fold with a fake low end). Under the hypotheses of Definition 12.2, both arms of S at 0 fold onto one walk c, with itineraries β_R (forward, on $[0, a-r]$) and β_L (backward), and there is a time U with $|U - \beta_R(a-r)| \le 1$ and $c(U) \le 1$.

Proof. Apply Proposition 6.5 and Lemma II.4.4 to S' (Lemma 12.3), whose forward arm has one more time than that of S. The itineraries restrict to S's arms. Let $U = \beta'_R(a - r + 1)$, the time at which the fake link is placed: $|c(U) - 0| \le 1$ gives $c(U) \le 1$, and the Lipschitz condition gives $|U - \beta_R(a - r)| \le 1$. □

Corollary 12.5 (prefix and suffix folds). For $r \le a$ with $E_a \cap D_0 \neq \emptyset$ there are a walk c_P, an itinerary β_P on $[0, a]$ with $\beta_P(r) = 0$, 1-Lipschitz, $|\rho(t) - c_P(\beta_P(t))| \le 1$ for $t \le a$, and a time U with $|U - \beta_P(a)| \le 1$, $c_P(U) \le 1$. Symmetrically, for $b \le s$ with $E_b \cap D_{k-1} \neq \emptyset$, reading the reversed chains and the reflected levels $k - 1 - \sigma$, there are c_Q, β_Q on $[b, n)$ with $\beta_Q(s) = 0$, $|(k - 1 - \sigma(t)) - c_Q(\beta_Q(t))| \le 1$ for $t \ge b$, and V with $|V - \beta_Q(b)| \le 1$, $c_Q(V) \le 1$.

Proof. Proposition 12.4 for the prefix; for the suffix, E reversed is crooked in D reversed (Lemma 2.6), whose level sequence is $k - 1 - \sigma$, and the link meeting the last link of D meets the first link of D reversed. □

13. The splice

Proposition 13.1 (the splice, on levels). Let $k \ge 6$, and let $r \le a < b \le s$ with $\rho(a) \le 1$ and $\sigma(b) \ge k - 2$. Given the prefix data (c_P, β_P, U) and suffix data (c_Q, β_Q, V) of Corollary 12.5, there is a 1-Lipschitz $\beta : \{0, \dots, n - 1\} \to \mathbb{N}$ with $\beta(r) = 0$, $\beta(t) \le \beta(s)$ for all t, and such that for every value v there is $w \in \mathbb{Z}$ with: every t with $\beta(t) = v$ has $\rho(t) \in [w, w + 2]$, or every such t has $\sigma(t) \in [w, w + 2]$.

Proof. Put $G = k - 5$ and define three kinds of index:

- *kept prefix*: $t \le a$ with $\beta_P(t) < U$;
- *kept suffix*: $t \ge b$ with $\beta_Q(t) < V$;
- *middle*: all others.

Set $\beta(t) = \beta_P(t)$ on the kept prefix; $\beta(t) = U + \mathrm{clamp}(\rho(t) - 2, 0, G)$ on middle indices $t < b$ and $U + \mathrm{clamp}(\sigma(t) - 2, 0, G)$ on middle indices $t \ge b$; and $\beta(t) = U + G + 1 + (V - 1 - \beta_Q(t))$ on the kept suffix.

The kept prefix is an initial segment. β_P is 1-Lipschitz with $\beta_P(r) = 0$; the set $\{t \le a : \beta_P(t) < U\}$ need not be an interval, but the first index at which β_P reaches U (if any, at or before a: $\beta_P(a) \ge U - 1$) cuts the prefix into a kept initial segment and a remainder that is middle. The formal proof takes the kept prefix to be that initial segment. Likewise for the suffix.

Lipschitz across the first cut. The last kept prefix index t_0 has $\beta_P(t_0) = U - 1$ and its successor $t_0 + 1$ has $\beta_P(t_0 + 1) = U$, so $|\rho(t_0 + 1) - c_P(U)| \le 1$ and $\rho(t_0 + 1) \le 2$; hence the middle value at $t_0 + 1$ is $U + 0$, adjacent to $U - 1$. This is the step for which Bing needs the end-link clause: it is $c_P(U) \le 1$ that puts the first middle link in the bottom group.

Lipschitz inside the middle. ρ and σ are 1-Lipschitz, $|\sigma(t + 1) - \rho(t)| \le 1$ handles the change of reading at b, and clamp is 1-Lipschitz.

Lipschitz across the second cut. Symmetric, using $c_Q(V) \le 1$: the first kept suffix index t_1 has $\beta_Q(t_1) = V - 1$, its predecessor has $\beta_Q = V$, so $|(k - 1 - \sigma(t_1 - 1)) - c_Q(V)| \le 1$ and $\sigma(t_1 - 1) \ge k - 3 = G + 2$; the middle value there is $U + G$, adjacent to $U + G + 1$.

The two kept parts are apart. $k \ge 6$ gives $b \ge a + 3$ (the levels rise from ≤ 1 at a to $\ge k - 2 \ge 4$ at b by unit steps), so the middle is nonempty and the kept suffix begins after the kept prefix ends.

Windows. A kept prefix block has $|\rho - c_P(u)| \le 1$: window $[c_P(u)-1, c_P(u)+1]$ in ρ. A middle group $U+g$ has its level (ρ before b, σ from b on) equal to $g+2$ when $0 < g < G$, at most 2 when $g = 0$, and at least $G+2 = k-3$ when $g = G$; each is a window of three levels inside $[0, k-1]$. A kept suffix block has $|(k-1-\sigma) - c_Q(v)| \le 1$: a window of three in σ. □

Lemma 13.2 (from a schedule to blocks). A 1-Lipschitz β with $\beta(r) = 0$ and $\beta \le \beta(s)$ is a block map onto $\{0, \dots, \beta(s)\}$ (it takes every value in between, by the discrete intermediate value property), so E/β is defined; if every block has ρ or σ in a window $[w, w+2]$, the block lies in $D_w \cup D_{w+1} \cup D_{w+2}$ (each link lies in $D_{\rho(t)}$ and in $D_{\sigma(t)}$).

Proposition 13.3 (the main case, $k \ge 6$). Theorem 10.1 holds when D has at least six links and the link meeting D_0 comes before the link meeting D_{k-1} ($r \le a < b \le s$).

Proof. Corollary 12.5, Proposition 13.1, Lemma 13.2. □

14. Small k, the other orientation, and the theorem

Lemma 14.1 (two windows). If $k \le 6$, every link of E lies in the first three links of D or in the last three. A block map alternating between the two windows along E — new block whenever the window changes — has every block in three consecutive links; with r in the first block and s in the last, adjusting the two ends, Theorem 10.1 holds. For $k = 3$ a single block serves.

Proof of Theorem 10.1. If the link meeting D_0 comes first, Proposition 13.3 ($k \ge 6$) or Lemma 14.1 ($k \le 6$). If the link meeting D_{k-1} comes first, apply the same to D reversed (E is crooked in it, Lemma 2.6; three consecutive links of D reversed are three consecutive links of D). □

15. Two-link basepoints

Theorem 13 (Part VII) needs E/β to be a *chain from R to S* (Definition 1.2). A point of a chain lies in at most two links, and they are adjacent. By Lemma 16.1 below, E/β is a chain from R to S exactly when every link of E containing R is in block 0 and every link containing S is in the last block. So the basepoints must be pairs.

Lemma 15.1 (block 0 of a fold). The fold of Part II from a basepoint i starts its walk at $\rho(i) - 1$ or $\rho(i) + 1$ (Proposition II.6.5), and its block 0 consists of the initial stretch of links whose level stays within one of the start. Hence a neighbour $i \pm 1$ of i lies in block 0 whenever $\rho(i \pm 1)$ is within one of the start.

Definition 15.2 (the two events). For a level function f on E, an index i and a bound $a \ge i$, EvUp(f, i, a) holds if some $t \le a$ has $f(t) = f(i) + 3$ with $f \ge f(i)$ strictly between i and t, or the fake link is at $f(i)+3$ and $f \ge f(i)$ on $[i, a]$; EvDown is the mirror with $f(i)-3$ and $f \le f(i)$.

EvUp(ρ, i, a) is what forbids the first direction "up first from $\rho(i) + 1$" in Lemma II.6.3 (an arm rises three levels before going below the basepoint level); EvDown(σ, i, a) forbids "down first from $\sigma(i) - 1$" for the fold on σ.

Proposition 15.3 (exclusion). Let (ρ, σ) be faithful (Lemma II.1.1), let lo, hi be adjacent indices with $\rho(hi) = \rho(lo) + 1$ and $\sigma(hi) = \rho(lo) + 1$, $lo \le a$, and suppose crookedness reads at the fake low end after a (as in Lemma 12.1). Then EvUp(ρ, lo, a) and EvDown(σ, hi, a) do not both hold.

Proof. Write $x = \rho(lo)$. The two events name a link t at ρ-level $x + 3$ reached from lo without going below x, and a link t' at σ-level $x - 2$ reached from hi without σ exceeding $x + 1$. If t and t'

lie on the same side of the pair, the first of them to be reached along E contradicts the level bound of the other event on the common stretch. If they lie on opposite sides, crookedness between the link at level $x+3$ and the link at level $x-2$ (five levels apart) returns, strictly between them, first a link within one of the far level and then one within one of the near level — a link at level $x+2$ or so on the side where the event bounds forbid it, and one at $x-1$ on the other; both contradict the bounds. Where the fake link plays the part of t or t', the crookedness clause with D_0 (Lemma 12.1) supplies the same two links. □

Proposition 15.4 (a fold whose block 0 holds both links). For an adjacent pair (r, r') and a later link a meeting D_0, there is a fold of the prefix $[0, a]$ from the pair — a walk, a 1-Lipschitz itinerary with value 0 at both r and r', reading the levels $\ell(t) \in [\rho(t), \sigma(t)]$ with $|\ell(t) - c(\beta(t))| \le 1$ — together with a time U within one of $\beta(a)$ with $c(U) \le 1$.

Proof. If $\rho(r) = \rho(r')$, either start of Lemma 15.1 puts both in block 0; Lemma II.6.3 provides one. Otherwise let lo be the lower link (level x) and hi the upper (level $x+1$); the adjacency of intervals gives $\sigma(hi) = x+1$. Either (a) the fold on ρ from lo started at $x+1$ is available — its block-0 window $[x, x+2]$ holds both links — or EvUp(ρ, lo, a) holds; either (b) the fold on σ from hi started at x is available — window $[x-1, x+1]$ holds $\sigma(lo) \in \{x, x+1\}$ and $\sigma(hi) = x+1$ — or EvDown(σ, hi, a) holds. By Proposition 15.3 one of (a), (b) is available. The fold on σ is Proposition 12.4 applied to the reflected sequence $k-1-\sigma$ with a fake *high* end. If $a = r$ (the link meeting D_0 is r itself) then $U = 0$ and r' enters block 0 through the middle grouping, since $\rho(r') \le 2$. □

Proof of Theorem 10.2. As Theorem 10.1, with Proposition 15.4 in place of Corollary 12.5 at both ends, the splice reading the level ℓ that each fold uses, and the splice reporting that links at prefix time 0 go to the first block and links at suffix time 0 to the last. For $k \le 5$, every link lies in the first three or the last three links of D, each basepoint pair lies in one window, and a two- or three-block schedule (the middle block being the high or low meeting link, which lies strictly between the pairs whenever it is needed) serves; for $k \ge 4$ the case $r' = s'$ is impossible, since crookedness places two links strictly between the low and high meeting links. □

Note. No four-link window was needed: the walk of Proposition 15.4 starts at a single level within one of both basepoint links, so block 0 is three links wide like every other block.

16. Chains from R to S

Lemma 16.1. E/β is a chain from R to S if and only if: R lies in some link of E, every link of E containing R has $\beta = 0$, S lies in some link of E, and every link of E containing S has $\beta = \max \beta$.

Proof. Definition 1.2 for E/β: $R \in$ block 0 means R is in some E_t with $\beta(t) = 0$; "R in no other block" means every $E_t \ni R$ has $\beta(t) = 0$. □

Corollary 16.2. If R lies in E_r alone and S in E_s alone, Theorem 10.1 gives a chain from R to S; if $R \in E_r \cap E_{r+1}$ or $S \in E_{s-1} \cap E_s$, Theorem 10.2 is needed and suffices.

Part IV. Theorem 10: every subcontinuum of M is indecomposable

17. Continua

Definition 17.1. A *continuum* is a compact connected set. A continuum K is *indecomposable* if whenever $K = A \cup B$ with A, B continua contained in K, one of A, B equals K. (A one-point set is a continuum and is indecomposable.)

Theorem 17.2 (Bing, Theorem 10). Let Δ be a description in X with continuum M. Every continuum $K \subseteq M$ is indecomposable.

No compactness of X is assumed: only closedness of the two pieces and connectedness of K and of each piece are used.

18. The separation lemma

Lemma 18.1 (a connected set inside a chain meets every link between two that it meets). Let E be a chain and $C \subseteq E^*$ a preconnected set. If C meets E_a and E_c with $a < b < c$, then C meets E_b.

Proof. Suppose $C \cap E_b = \emptyset$. Put $U = \bigcup_{t<b} E_t$ and $V = \bigcup_{t>b} E_t$, open sets. $C \subseteq U \cup V$ since $C \subseteq E^*$ misses E_b. $U \cap V = \emptyset$: a link with index $< b$ and one with index $> b$ are at index distance ≥ 2, so disjoint. C meets U (at E_a) and V (at E_c). This contradicts preconnectedness of C. □

Lemma 18.2 (mesh). If every link of D is bounded with diameter at most ε, points of links at index distance d are within $(d+1)\varepsilon$ of each other. *Proof.* Induction along the intermediate links, each step through a common point of two adjacent links. □

19. Proof of Theorem 17.2

Let $K \subseteq M$ be a continuum and $K = A \cup B$ with A, B continua inside K, $A \neq K$, $B \neq K$. Pick $p \in K \setminus A$ and $q \in K \setminus B$. Since A and B are closed there is $\delta > 0$ with the ball $B(p, \delta)$ missing A and $B(q, \delta)$ missing B. Choose a level j with $\varepsilon_j < \delta/3$. Since $p, q \in M \subseteq (D^{(j)})^*$, let $p \in D^{(j)}_h$ and $q \in D^{(j)}_k$.

The two links are far apart. If $|h - k| \leq 2$ then $\operatorname{dist}(p, q) \leq 3\varepsilon_j < \delta$ by Lemma 18.2, so $q \in B(p, \delta)$; but $q \in K = A \cup B$, $q \notin B$, so $q \in A$, and A misses $B(p, \delta)$: contradiction. Say $h + 2 < k$ (the other case is symmetric).

The neighbouring links miss the wrong piece. Every point of $D^{(j)}_{h+1}$ is within $2\varepsilon_j < \delta$ of p (Lemma 18.2), so $D^{(j)}_{h+1} \cap A = \emptyset$; likewise $D^{(j)}_{k-1} \cap B = \emptyset$.

Crookedness one level down. p and q lie in $M \subseteq (D^{(j+1)})^*$; let $E = D^{(j+1)}$, $p \in E_i$, $q \in E_{i'}$. E_i meets $D^{(j)}_h$ and $E_{i'}$ meets $D^{(j)}_k$, so by crookedness of E in $D^{(j)}$ there are r, s with $\operatorname{Between}(i, r, s, i')$, $E_r \subseteq D^{(j)}_{k-1}$ and $E_s \subseteq D^{(j)}_{h+1}$. In particular $E_r \cap B = \emptyset$ and $E_s \cap A = \emptyset$.

The separation. K is connected, lies in E^*, and meets E_i (at p) and $E_{i'}$ (at q); r lies between, so K meets E_r (Lemma 18.1). A point of $K \cap E_r$ is not in B, so it is in A. Thus A meets E_r and $E_{i'}$ (at $q \in A$). A is connected, lies in E^*, and s lies between r and i'; so A meets E_s (Lemma 18.1). But $E_s \cap A = \emptyset$. □

Note. Bing's proof uses the distance $2/j$ and the subchains $D_j(h,k)$, $D_{j+1}(u,v)$ determined by P and Q; the formal proof works with the two links holding p and q directly and needs no subchain. The step "H is not a continuum since $d(j+1)_r$ and $d(j+1)_t$ contain points of H and $d(j+1)_s$ does not" is Lemma 18.1.

Corollary 19.1 (Part VII uses it). M itself is a continuum when X is compact (Proposition 29.4), and hence indecomposable.

Part V. Theorems 5 and 6: pattern following

20. Statements

Fix a description Δ with chains $D^{(0)}, D^{(1)}, \dots$ in a compact metric space X.

Theorem 20.1 (Bing, Theorem 6). Let $P \neq Q$. For every pattern $x : \{0, \dots, N-1\} \to \{0, \dots, n_0 - 1\}$ (Definition 1.6: $x(0) = 0$, $x(N-1) = n_0 - 1$, unit steps) there are $j > 0$ and a chain F from P to Q with N links which is a consolidation of $D^{(j)}$ and follows x in $D^{(0)}$.

Applied to every tail $\Delta_{\geq i}$, this is the hypothesis under which Theorem 12 was first proved (Part VI): *pattern following*, for every i and every pattern x in $D^{(i)}$, some deeper level has a consolidation from P to Q following x in $D^{(i)}$.

Proposition 20.2 ($P \neq Q$ is necessary). There is a description with $P = Q$ for which Theorem 20.1 fails: on the one-point space, every level a single link (the whole space), $\varepsilon_i = 1/(i+1)$. The pattern $0, 0$ of length two is a pattern in $D^{(0)}$; no chain of two links is a chain from P to P, since P would have to lie in the first link only and in the last link only.

Note. Bing's Theorem 6 does not state $P \neq Q$; it is implicit in "chains from P to Q" of more than one link. Proposition 20.2 is a kernel-checked refutation of the statement without it.

Theorem 20.3 (Bing, Theorem 5; statement). Let $x : \{0, \dots, N-1\} \to \{0, \dots, n_0 - 1\}$ be a pattern *from h to k*: $x(0) = h$, $x(N-1) = k$, $h \leq x(t) \leq k$, unit steps. Let $u \leq v$ be indices of $D^{(1)}$ such that every link $D^{(1)}_t$ with $u \leq t \leq v$ lies in some $D^{(0)}_s$ with $h \leq s \leq k$, $\overline{D^{(1)}_u} \subseteq D^{(0)}_h$, $\overline{D^{(1)}_v} \subseteq D^{(0)}_k$, and these two closures are disjoint. Then for every w there are $j > w$ and a chain F with N links following x in $D^{(0)}$, such that F is a consolidation of the set of links of $D^{(j)}$ lying in some $D^{(1)}_t$ with $u \leq t \leq v$ (every such link lies in a link of F, and every link of F is a union of such links), and no interior link of F meets $D^{(1)}_u \cup D^{(1)}_v$.

Theorem 20.1 is not derived from Theorem 20.3 here; both are derived from one combinatorial statement (§21). Theorem 20.3 is included for completeness; nothing in Parts VI–VII uses it.

21. The combinatorial core

The setting is two integer readings of one time axis.

Definition 21.1. A *fine reading* is $\sigma : \mathbb{N} \to \mathbb{Z}$ with $|\sigma(f+1) - \sigma(f)| \leq 1$, *crooked up to T*: whenever $f < f' \leq T$ and $|\sigma(f) - \sigma(f')| \geq 3$, there are $f < r < s < f'$ with $|\sigma(r) - \sigma(f')| \leq 2$ and $|\sigma(s) - \sigma(f)| \leq 2$.

(Geometrically, $\sigma(f)$ is the index of a link of a reference chain R containing link f of a fine chain F; crookedness of F in R gives this, with slack 2 because the witnesses are within one link of the target and the reading may be off by one.)

Definition 21.2. A *coarse reading* $\tau : \mathbb{Z} \to \mathbb{Z}$ is *w-slow* if $|p - p'| \le w$ implies $|\tau(p) - \tau(p')| \le 1$.

(Geometrically: $\tau(p)$ is a link of the coarse chain containing every link of R within w positions of p — a thick reading, Lemma 3.5.)

Definition 21.3. A value v is *allowed at time f with radius d* if some position p with $|p - \sigma(f)| \le d$ has $\tau(p) = v$. A *lift* of the pattern segment $x|_{[p,q]}$ over the time interval $[a, b]$ from index i to index j is a map $\varphi : [a, b] \to [p, q]$ with $\varphi(a) = i$, $\varphi(b) = j$, unit steps, and $x(\varphi(f))$ allowed at f with radius w for every f.

Definition 21.4 (the induction hypothesis Good**).** Given σ, T, a budget B, τ, x, indices $p \le q$, times $a \le b \le T$ and a radius e: Good(p, q, a, b, e) holds if $q - p \le b - a$; x has unit steps on $[p, q]$ with $x(p)$ its minimum and $x(q)$ its maximum there; $\tau(\sigma(f)) \in [x(p), x(q)]$ for $f \in [a, b]$; $x(p)$ is allowed at a with radius e and $x(q)$ at b with radius e (the *anchors*); and $e + 2(q - p) \le B$.

Theorem 21.5 (the lift exists). Let σ be crooked up to T, $2B + 10 \le w$, τ w-slow, and Good(p, q, a, b, e). Then there is a lift of $x|_{[p,q]}$ over $[a, b]$ from p to q.

Proof. Strong induction on $b - a$. Write $\ell(f) = \tau(\sigma(f))$; it has unit steps ($\sigma$ does and τ is slow).

Case $x(q) = x(p)$ (constant pattern). Every $\ell(f)$ equals the common value, which is allowed at every time with radius 0; ramp φ from p to q over the first $q - p$ steps (possible since $q - p \le b - a$) and hold.

Case $x(p + 1) = x(p)$ (a plateau at the start; Bing's (P4)). Lift $x|_{[p+1,q]}$ over $[a + 1, b]$ by induction: Good$(p + 1, q, a + 1, b, e + 1)$ holds — the anchor at $a + 1$ is $x(p + 1) = x(p)$, allowed at a with radius e, hence at $a+1$ with radius $e+1$ since σ moves by one; the budget $e+1+2(q-p-1) \le B$. Prepend $\varphi(a) = p$. The symmetric case, a plateau at the end, is the same with time reversed.

Case $x(q) = x(p) + 1$ with no plateau at either end. Two adjacent values; there is a time in $[a, b]$ near which both are allowed (the reading ℓ passes from $x(p)$ to $x(q)$, and slowness gives a position reading each value within w of it); ramp to the crossing index there. Crookedness is not used.

Case $x(q) \ge x(p) + 2$, $x(p + 1) = x(p) + 1$, and x returns to $x(p)$ at some $i \in (p, q]$ (Bing's (P5)). Let i be the first return and $t \in (p, i)$ a position of the maximum m of x on $[p, i]$. Let f_1 be the first time in $[a, b]$ with $\ell(f_1) > m$ — it exists, since $\ell(b) = x(q) \ge$ ... the formal proof handles the case where the pattern's peak m is not exceeded by first lifting to b directly. Crookedness of σ between a and f_1 (the readings differ by at least $m + 1 - x(p) \ge 3$ levels of τ, hence the positions differ by more than $w \ge 3$ ...) returns times $a < r < s < f_1$ with $\sigma(r)$ within 2 of $\sigma(f_1)$ and $\sigma(s)$ within 2 of $\sigma(a)$: the reading rises to m near r and comes back to $x(p)$ near s. Split: lift $x|_{[p,t]}$ over $[a, r]$ (an ascent), $x|_{[t,i]}$ over $[r, s]$ (a descent: apply the induction to the reflected pattern and negated readings — negation and reflection preserve all the hypotheses), and $x|_{[i,q]}$ over $[s, b]$; glue at r and s, where the anchors match with radius at most 2, within budget. Each piece is shorter in time than $[a, b]$.

Case $x(q) \ge x(p) + 2$, $x(p + 1) = x(p) + 1$, and x never returns to $x(p)$ (Bing's (P6)). Let f_0 be the last time in $[a, b]$ with $\ell(f_0) = x(p)$. Lift $x|_{[p+1,q]}$ over $[f_0 + 1, b]$ by induction (the reading is $\ge x(p) + 1$ there). Before f_0 the reading makes excursions above $x(p)$ and returns; lift each

excursion by splitting it at its highest time and lifting up then down (time reversal), staying at index p or $p+1$ as the reading permits; the anchor at f_0 is $x(p)$ with radius 0.

In every case the pieces are shorter time intervals, and Good holds for them with the budget accounting above. □

Corollary 21.6 (the lift over a whole interval). Let σ be crooked up to T, x a pattern of length N with $x(0)$ its minimum and $x(N-1)$ its maximum, $w \geq 4N+10$, τ w-slow, $\ell = \tau \circ \sigma$ with values in $[x(0), x(N-1)]$ on $[0,T]$, $\ell(0) = x(0)$, $\ell(T) = x(N-1)$, $T \geq N-1$. Then there is a lift $\varphi : [0,T] \to [0, N-1]$ from 0 to $N-1$ with unit steps and $x(\varphi(f))$ allowed at f with radius w.

Proof. Theorem 21.5 with $B = 2(N-1)$, $e = 0$, $p = 0$, $q = N-1$, $a = 0$, $b = T$. □

Note. The anchors with a radius budget are what Bing's elided step in (P4) — "we omit the details of the determination of j and E" — turns out to require: dropping the first link of the pattern costs one unit of radius at the anchor, and the budget bounds the total cost by twice the pattern length. The window $w \geq 4N+10$, as long as the pattern, is needed for this; Bing's fixed "five-linked subchains" would not suffice on this route.

22. Theorem 6 from the core

Lemma 22.1 (a lift is a block map). A unit-step $\varphi : [0,T] \to [0, N-1]$ with $\varphi(0) = 0$, $\varphi(T) = N-1$ is a block map onto $\{0, \dots, N-1\}$.

Proposition 22.2 (the consolidation). Let $F = D^{(m+1)}$, $R = D^{(m)}$, and let τ be a thick reading of R in $D^{(0)}$ with radius w (Lemma 3.5) and σ a reading of F in R (each link of F inside link $\sigma(f)$ of R). If φ is a lift as in Corollary 21.6, then F/φ is a chain from P to Q, a consolidation of F, following x in $D^{(0)}$.

Proof. F/φ is a block consolidation (Lemma 22.1, Lemma 1.5). Its link u is the union of the links F_f with $\varphi(f) = u$; each such $F_f \subseteq R_{\sigma(f)}$, and $x(u)$ is allowed at f with radius w: some position p within w of $\sigma(f)$ has $\tau(p) = x(u)$, and thickness puts $R_{\sigma(f)}$ inside $D^{(0)}_{\tau(p)} = D^{(0)}_{x(u)}$. So F/φ follows x. It is a chain from P to Q: P lies in F_0 alone and $\varphi(0) = 0$; Q in F_T alone and $\varphi(T) = N-1$. □

Proof of Theorem 20.1. Put $w = 4N+10$. Choose m with a w-thick reading of $D^{(m)}$ in $D^{(0)}$ (Lemma 3.5) and with $D^{(m+1)}$ having at least N links: the number of links of $D^{(i)}$ tends to infinity when $P \neq Q$, since a chain from P to Q of mesh ε has at least $\operatorname{dist}(P,Q)/\varepsilon$ links (Lemma 18.2). The reading σ of $D^{(m+1)}$ in $D^{(m)}$ is crooked up to $T = n_{m+1} - 1$ (crookedness of $D^{(m+1)}$ in $D^{(m)}$, read on indices with slack 2). The reading $\ell = \tau \circ \sigma$ runs from 0 at P's link to $n_0 - 1$ at Q's link and stays in between. Corollary 21.6 and Proposition 22.2 give F at level $j = m+1 > 0$. □

23. Theorem 5 from the core

The proof of Theorem 20.3 runs the same lift with a coarse reading that has two *end values* and then places the links of $D^{(j)}$ in $D^{(1)}(u,v)$ run by run.

The reading. Let $W = 4(N+2)+10$ and choose m deep enough that (Lemma 3.5 and its variant for closures inside open sets): links of $D^{(m)}$ within W positions of a link meeting $D^{(1)}_u$ lie in $D^{(0)}_h$; those within W of a link meeting $D^{(1)}_v$ lie in $D^{(0)}_k$; the two kinds of position are more than $2W$

apart (the closures of $D_u^{(1)}$ and $D_v^{(1)}$ are disjoint compact sets); and $D^{(m)}$ has a W-thick reading τ_1 in $D^{(1)}$. Define τ on positions of $D^{(m)}$: $h-1$ within W of a link meeting $D_u^{(1)}$; $k+1$ within W of one meeting $D_v^{(1)}$; h (resp. k) in a ring of width W around each of those zones; and elsewhere the index of a link of $D^{(0)}(h,k)$ containing $D^{(1)}_{\tau_1(p)}$.

The pattern. $x^{\text{ext}} = (h-1, x(0), \dots, x(N-1), k+1)$, of length $N+2$, with minimum $h-1$ at the start and maximum $k+1$ at the end.

Soundness. For a link f of $D^{(j)}$ ($j = m+1$) lying in $D^{(1)}(u,v)$: if $h-1$ is allowed at f then $D_f^{(j)} \subseteq D_h^{(0)}$; if $k+1$ then $\subseteq D_k^{(0)}$; if $c \in [h,k]$ then $D_f^{(j)} \subseteq D_c^{(0)}$ and $D_f^{(j)}$ misses $D_u^{(1)} \cup D_v^{(1)}$.

Forcing. Near a link meeting $D_u^{(1)}$ only $h-1$ is read, so a lift must be at index 0 there; near $D_v^{(1)}$, at index $N+1$.

Runs. The links of $D^{(j)}$ lying in $D^{(1)}(u,v)$ form maximal runs of consecutive indices. Each run begins and ends at a link inside $D_u^{(1)}$ or $D_v^{(1)}$ (or at an end of $D^{(j)}$, which holds P or Q). A run from u's side to v's side is lifted by Corollary 21.6; from v to u, by the same with time reversed; an *excursion* — a run leaving $D_u^{(1)}$ and returning to it without reaching $D_v^{(1)}$ — is lifted by splitting at its highest reading and lifting up and back down. Excursions are not in Bing's text; at a deep level they exist, and F must place their links.

Assembly. Shifting every lift down by one and clamping to $[0, N-1]$ assigns each link of $D^{(j)}$ in $D^{(1)}(u,v)$ a block. Some run crosses from u to v (an intermediate value argument on the reading of $D^{(j)}$ in $D^{(1)}$), so every block is nonempty and consecutive blocks are adjacent; the unions of the blocks form a chain F with N links. Soundness gives the three conclusions: F follows x in $D^{(0)}$; F is a consolidation of the links in $D^{(1)}(u,v)$; interior links of F (indices $1, \dots, N-2$, which read values in $[h,k]$) miss $D_u^{(1)} \cup D_v^{(1)}$. And $j = m+1 > w$ for m large. □

Part VI. Theorems 11 and 12: matching covers, and the uniqueness of the described continuum

24. Statements

Throughout this Part, $M_1 \subseteq X_1$ and $M_2 \subseteq X_2$ are compact subsets of metric spaces, and $(\varepsilon_i)_{i \geq 0}$ is a sequence of positive reals with $\sum_i \varepsilon_i < \infty$.

Definition 24.1 (matching covers). A system of *matching covers* of M_1, M_2 with constants ε_i consists of, for every level $i \geq 0$, a number N_i of indices and two families of open sets $x^1_{i,k} \subseteq X_1$, $x^2_{i,k} \subseteq X_2$ ($0 \leq k < N_i$), indexed by the *same* index set at each level, such that for $n = 1, 2$ and every i:

1. (cover) $M_n \subseteq \bigcup_k x^n_{i,k}$;
2. (meet) every $x^n_{i,k}$ meets M_n;
3. (diameter) every $x^n_{i,k}$ is bounded, with diam $x^n_{i,k} \leq \varepsilon_i$;
4. (match) if $x^1_{i+1,j}$ meets $x^1_{i,k}$, then some point of $x^2_{i+1,j}$ is within ε_i of some point of $x^2_{i,k}$; and the same with the superscripts $1, 2$ exchanged.

An index sequence $a = (a_i)_i$ with $a_i < N_i$ *tracks* a point $p \in X_1$ on side 1 if $p \in x^1_{i,a_i}$ for every i, and tracks $q \in X_2$ on side 2 if $q \in x^2_{i,a_i}$ for every i.

Note. The condition is symmetric in the two sides; exchanging the superscripts gives a system of matching covers of M_2, M_1, and every one-sided statement below has a mirror image obtained this way. Boundedness in (3) is stated because the diameter of an unbounded set is 0 by the convention of the formal library.

Theorem 24.2 (Bing, Theorem 11; with tracking). Given matching covers of compact M_1, M_2 with summable positive constants ε_i, there is a homeomorphism $T : M_1 \to M_2$ such that whenever one index sequence a tracks $p \in M_1$ on side 1 and $q \in M_2$ on side 2, $T(p) = q$.

Bing's statement (p. 738) has conditions (1)–(4) in this form and concludes only that M_1 and M_2 are homeomorphic. The tracking clause is what Theorem 12 uses to carry the end points across; it comes out of Bing's construction of T without change.

Theorem 24.3 (Bing, Theorem 12). Let Δ_1 be a description from P_1 to Q_1 in a compact metric space X_1 and Δ_2 a description from P_2 to Q_2 in a compact metric space X_2 (Definition 3.1), with $P_1 \neq Q_1$ and $P_2 \neq Q_2$. Then there is a homeomorphism $T : M_1 \to M_2$ of the described continua with $T(P_1) = P_2$ and $T(Q_1) = Q_2$.

The hypotheses $P_n \neq Q_n$ are not in Bing's statement; they are needed for Theorem 20.1 (Proposition 20.2) and hence here. Bing's proof of Theorem 12 (pp. 738–739) constructs two sequences of chains, one in each space, with the same numbers of links and the same patterns, and applies Theorem 11 to them. The proof below follows that plan exactly, with Theorem 20.1 supplying the chains; §26 packages one level of the construction as a *rung* and §27 runs the induction and the bookkeeping of the mesh.

25. Fattenings and the map T

Fix matching covers as in Definition 24.1. Since $\sum \varepsilon_i$ converges, the tails are finite.

Definition 25.1 (radii and mesh). For $i \geq 0$,

$$\operatorname{tail}(i) = \sum_{l \geq 0} \varepsilon_{i+l}, \qquad \operatorname{rad}(i) = \varepsilon_i + 2\operatorname{tail}(i+1), \qquad \operatorname{mesh}(i) = \varepsilon_i + 2\operatorname{rad}(i).$$

For a set S and $\delta > 0$, the *fattening* S^δ is the open δ-neighbourhood $\{z : \operatorname{dist}(z, S) < \delta\}$; a point lies in S^δ if and only if it is within δ of a point of S. The *fattened elements* are

$$y^n_{i,k} = \left(x^n_{i,k}\right)^{\operatorname{rad}(i)}, \qquad n = 1, 2.$$

These are Bing's $y(n,i)_k$. The following facts about the constants are immediate: $\operatorname{tail}(i) = \varepsilon_i + \operatorname{tail}(i+1)$; $\operatorname{rad}(i) > 0$; $\operatorname{rad}(i) = \operatorname{rad}(i+1) + \varepsilon_{i+1} + \varepsilon_i$; and $\operatorname{tail}(i) \to 0$, hence $\operatorname{rad}(i) \to 0$ and $\operatorname{mesh}(i) \to 0$. Since $x^n_{i,k}$ has diameter at most ε_i and a δ-fattening raises the diameter by at most 2δ, every $y^n_{i,k}$ has diameter at most $\operatorname{mesh}(i)$, and so does its closure. In particular:

Lemma 25.2. Two points of $\overline{y^n_{i,k}}$ are within $\operatorname{mesh}(i)$ of each other. For every $\delta > 0$ and every m there is $j > m$ with $\operatorname{mesh}(j) < \delta$. Two points q, q' with $\operatorname{dist}(q, q') \leq \operatorname{mesh}(i)$ for every i are equal.

Lemma 25.3 (the nesting computation). Let A, B be sets in a metric space with B bounded, diam $B \le \varepsilon_{i+1}$, and suppose some point $p \in B$ is within ε_i of some point $q \in A$. Then

$$\overline{B^{\mathrm{rad}(i+1)}} \subseteq A^{\mathrm{rad}(i)}.$$

Proof. Let z be in the closure and put $\eta = \varepsilon_i - \mathrm{dist}(p,q) > 0$. Choose $z' \in B^{\mathrm{rad}(i+1)}$ with $\mathrm{dist}(z,z') < \eta$ and $b' \in B$ with $\mathrm{dist}(z',b') < \mathrm{rad}(i+1)$. Then

$$\mathrm{dist}(z,q) \le \mathrm{dist}(z,z') + \mathrm{dist}(z',b') + \mathrm{dist}(b',p) + \mathrm{dist}(p,q) < \eta + \mathrm{rad}(i+1) + \varepsilon_{i+1} + \mathrm{dist}(p,q) = \mathrm{rad}(i+1) + \varepsilon_{i+1} + \varepsilon$$

so $z \in A^{\mathrm{rad}(i)}$. □

This is the one computation in Bing's proof; everything else is the following bookkeeping.

Lemma 25.4 (step 1: consecutive levels). If $x^1_{i+1,s}$ meets $x^1_{i,r}$ then $\overline{y^2_{i+1,s}} \subseteq y^2_{i,r}$. Mirror: if $x^2_{i+1,s}$ meets $x^2_{i,r}$ then $\overline{y^1_{i+1,s}} \subseteq y^1_{i,r}$.

Proof. Condition (4) gives $p \in x^2_{i+1,s}$, $q \in x^2_{i,r}$ with $\mathrm{dist}(p,q) < \varepsilon_i$; apply Lemma 25.3 with $B = x^2_{i+1,s}$, $A = x^2_{i,r}$. □

Lemma 25.5 (step 2: distant levels through a point of M). Let $i < j$, and suppose $x^1_{i,r}$ and $x^1_{j,s}$ have a common point $c \in M_1$. Then $\overline{y^2_{j,s}} \subseteq y^2_{i,r}$. Mirror image likewise.

Proof. Induction on j. If $j = i+1$ this is Lemma 25.4. If $j = m+1$ with $i < m$: by (1) at level m, $c \in x^1_{m,t}$ for some t; Lemma 25.4 gives $\overline{y^2_{m+1,s}} \subseteq y^2_{m,t}$, and the induction hypothesis (with $c \in x^1_{i,r} \cap x^1_{m,t}$) gives $\overline{y^2_{m,t}} \subseteq y^2_{i,r}$. □

Note. Bing's step 2 passes through the point c of M_1 in exactly this way; the point must be in M_1 so that (1) applies at the intermediate levels. Steps 1 and 2 are the only places where the sides interact.

Lemma 25.6 (step 4: a tracking sequence has a limit point). Let a be an index sequence such that $x^1_{i+1,a_{i+1}}$ meets x^1_{i,a_i} for every i. Then there is a point $q \in M_2$ with $q \in \overline{y^2_{i,a_i}}$ for every i.

Proof. The sets $t_i = \overline{y^2_{i,a_i}} \cap M_2$ are closed, nonempty (by (2), x^2_{i,a_i} meets M_2, and $x^2_{i,a_i} \subseteq y^2_{i,a_i}$), nested ($t_{i+1} \subseteq t_i$ by Lemma 25.4) and lie in the compact set M_2; so they have a common point. □

Definition 25.7 (the map). For $p \in M_1$ choose, by (1), an index sequence a^p with $p \in x^1_{i,a^p_i}$ for every i; consecutive elements meet (in p), so Lemma 25.6 applies. Let $T(p)$ be a point of M_2 lying in $\overline{y^2_{i,a^p_i}}$ for every i.

Lemma 25.8 (step 5: independence of the choice). For every $p \in M_1$, every level j and every index r with $p \in x^1_{j,r}$: $T(p) \in y^2_{j,r}$.

Proof. $p \in x^1_{j+1,a^p_{j+1}} \cap x^1_{j,r}$, so Lemma 25.4 gives $\overline{y^2_{j+1,a^p_{j+1}}} \subseteq y^2_{j,r}$, and $T(p)$ lies in the left side. □

So $T(p)$ lies in every fattened level-j element on side 2 whose partner on side 1 contains p, whichever tracking sequence was chosen. The remaining steps use only Lemma 25.8 and Lemma 25.2.

Proposition 25.9 (step 6: continuity). $T : M_1 \to M_2$ is continuous.

Proof. Let $p \in M_1$ and $\rho > 0$. Choose j with $\mathrm{mesh}(j) < \rho$ (Lemma 25.2), then r with $p \in x^1_{j,r}$ (by (1)), then $\delta > 0$ with the δ-ball about p inside the open set $x^1_{j,r}$. For $p' \in M_1$ with $\mathrm{dist}(p, p') < \delta$, both $T(p)$ and $T(p')$ lie in $y^2_{j,r}$ (Lemma 25.8), so $\mathrm{dist}(T(p), T(p')) \le \mathrm{mesh}(j) < \rho$. □

Proposition 25.10 (step 8: injectivity). T is injective.

Proof. Let $p_1 \neq p_2$ in M_1 and suppose $T(p_1) = T(p_2) = Q$. Choose k with $\mathrm{mesh}(k) < \mathrm{dist}(p_1, p_2)$; by (1) choose r with $Q \in x^2_{k,r}$, and $\rho > 0$ with the ρ-ball about Q inside $x^2_{k,r}$; choose $j > k$ with $\mathrm{mesh}(j) < \rho$; by (1) choose u, v with $p_1 \in x^1_{j,u}$, $p_2 \in x^1_{j,v}$. By Lemma 25.8, $Q \in y^2_{j,u} \cap y^2_{j,v}$.

Claim: for any index w at level j with $Q \in y^2_{j,w}$, $\overline{y^1_{j,w}} \subseteq y^1_{k,r}$. By (2) pick $c \in x^2_{j,w} \cap M_2$; then $c \in y^2_{j,w}$, so $\mathrm{dist}(c, Q) \le \mathrm{mesh}(j) < \rho$ and $c \in x^2_{k,r}$. Thus $x^2_{k,r}$ and $x^2_{j,w}$ share the point c of M_2, and the mirror of Lemma 25.5 gives the claim.

Applying the claim to $w = u$ and $w = v$: $p_1 \in x^1_{j,u} \subseteq y^1_{j,u}$ gives $p_1 \in y^1_{k,r}$, and likewise $p_2 \in y^1_{k,r}$. So $\mathrm{dist}(p_1, p_2) \le \mathrm{mesh}(k) < \mathrm{dist}(p_1, p_2)$, a contradiction. □

Proposition 25.11 (step 7: surjectivity). T is onto M_2.

Proof. $T(M_1)$ is compact (Proposition 25.9), hence closed in M_2; it suffices to show it is dense. Let $q \in M_2$ and $\rho > 0$. Choose j with $\mathrm{mesh}(j) < \rho$, then r with $q \in x^2_{j,r}$ (by (1)), then, by (2), $p \in x^1_{j,r} \cap M_1$. By Lemma 25.8, $T(p) \in y^2_{j,r}$; also $q \in y^2_{j,r}$; so $\mathrm{dist}(T(p), q) \le \mathrm{mesh}(j) < \rho$. □

Proof of Theorem 24.2. T is a continuous bijection from the compact space M_1 onto the Hausdorff space M_2 (Propositions 25.9–25.11), hence a homeomorphism. For the tracking clause, let a track p on side 1 and q on side 2. For every i, $T(p) \in y^2_{i,a_i}$ (Lemma 25.8, since $p \in x^1_{i,a_i}$) and $q \in x^2_{i,a_i} \subseteq y^2_{i,a_i}$, so $\mathrm{dist}(T(p), q) \le \mathrm{mesh}(i)$; by Lemma 25.2, $T(p) = q$. □

Note. Bing's steps are numbered 1–8 on pp. 738–739; step 3 there is the observation that T is well defined, which is Lemma 25.8 here, and his step 8 (injectivity) is proved before his step 7 (surjectivity) above because surjectivity uses compactness of the range. Nothing about chains, patterns or crookedness enters this section.

26. The alternating construction

Now let Δ_1, Δ_2 be descriptions as in Theorem 24.3, with chains $D^{(i)}_1$ (with $n_{1,i}$ links) and $D^{(i)}_2$ (with $n_{2,i}$ links), constants $\varepsilon_{1,i}$, $\varepsilon_{2,i}$, and continua M_1, M_2.

Recall from Theorem 20.1 and the paragraph after it that Δ_n has *pattern following*: for every level i and every pattern $x : \{0, \dots, N-1\} \to \{0, \dots, n_{n,i} - 1\}$ (Definition 1.6) there are $j > i$ and a chain E from P_n to Q_n with N links which is a consolidation of $D^{(j)}_n$ and follows x in $D^{(i)}_n$. This is Theorem 20.1 applied to the tail $\Delta_{n,\ge i}$ (Definition 3.3), which is a description from P_n to Q_n with $P_n \neq Q_n$.

Lemma 26.1 (pattern matching). Let Y_1 follow the pattern x in Z_1 (chains in X_1) and Y_2 follow the same x in Z_2 (chains in X_2), where Y_1, Y_2 have N links and Z_1, Z_2 have m links, and suppose every link of Z_2 is bounded with diameter $< \varepsilon$. If link j of Y_1 meets link k of Z_1, then some point of link j of Y_2 is within ε of some point of link k of Z_2.

Proof. Link j of Y_1 lies in link $x(j)$ of Z_1; since it meets link k, links $x(j)$ and k of Z_1 meet, so $|x(j) - k| \le 1$; hence links $x(j)$ and k of Z_2 meet, in a point q say. Any point p of link j of Y_2 lies

in link $x(j)$ of Z_2, so $\operatorname{dist}(p,q) \le \operatorname{diam}(Z_2)_{x(j)} < \varepsilon$. $\square$

This is condition (4) of Definition 24.1, for two pairs of chains following the same pattern, with ε any bound exceeding the mesh of the coarser pair. It is the only place where "same pattern" is used.

Lemma 26.2 (mesh is inherited). If Y follows a pattern in Z, or merely refines Z, and every link of Z is bounded with diameter $\le \varepsilon$, then every link of Y has diameter $\le \varepsilon$.

Lemma 26.3 (a refinement between chains from P to Q is a pattern). Let Y (with N links) and Z (with m links) be chains from P to Q in the same space, with Y refining Z. Choose for each link t of Y a link $x(t)$ of Z containing it. Then x is a pattern in the sense of Definition 1.6 ($x(0) = 0$, $x(N-1) = m-1$, unit steps) and Y follows x in Z.

Proof. Y follows x by construction, so x has unit steps (Lemma 1.7). $P \in Y_0 \subseteq Z_{x(0)}$ and P lies in no link of Z but Z_0, so $x(0) = 0$; likewise $x(N-1) = m-1$ at Q. $\square$

The construction alternates between the two sides. At each stage one side is a *level* of its description — some $D_n^{(c)}$ itself — and therefore has pattern following available for patterns *in it*; the other side is a consolidation of a level, which need not have that property. The side that is a level is refined by a deeper level of small mesh (Lemma 26.3 turns the refinement into a pattern), and the other side follows that pattern by pattern following, becoming a consolidation of a level; the roles then reverse. The following definition names the property that lets a side be followed.

Definition 26.4 (can follow). For a description Δ from P to Q, a level ℓ and a chain C with N links, say Δ *can follow C above ℓ* if for every N' and every pattern $x : \{0,\dots,N'-1\} \to \{0,\dots,N-1\}$ there are $a > \ell$ and a chain E from P to Q with N' links, a consolidation of $D^{(a)}$, following x in C.

Pattern following says exactly that Δ can follow $D^{(i)}$ above i, for every i.

Lemma 26.5 (basic facts about consolidations of levels). Let C be a consolidation of $D^{(\ell)}$ for a description Δ in a compact space. Then $M \subseteq C^*$, and every link of C meets M.

Proof. $M \subseteq (D^{(\ell)})^* \subseteq C^*$ (Lemma 3.2 and refinement). A link of C is a nonempty union of links of $D^{(\ell)}$, each of which meets M (Lemma 3.2). $\square$

Lemma 26.6 (a deeper level refines a consolidation). If C is a consolidation of $D^{(\ell)}$ and $\delta > 0$, there is $c \ge \ell$ with $D^{(c)}$ refining C and $\varepsilon_c \le \delta$.

Proof. $D^{(c)}$ refines $D^{(\ell)}$ for $c \ge \ell$ (each level refines the previous one, by Definition 3.1(3)), which refines C; and $\varepsilon_c \to 0$. $\square$

Proposition 26.7 (one half-step). Let Δ_A (from P_A to Q_A in X_A) and Δ_B (from P_B to Q_B in X_B) be descriptions, and let C_A, C_B be chains with the same number N of links such that Δ_A can follow C_A above ℓ_A, and C_B is a chain from P_B to Q_B and a consolidation of $D_B^{(\ell_B)}$. Let $\delta > 0$. Then there are: a level $c \ge \ell_B$ with $\varepsilon_{B,c} \le \delta$; a level $\ell'_A > \ell_A$; a chain C'_A from P_A to Q_A with $n_{B,c}$ links which is a consolidation of $D_A^{(\ell'_A)}$; and a pattern $x : \{0,\dots,n_{B,c}-1\} \to \{0,\dots,N-1\}$ such that $D_B^{(c)}$ follows x in C_B and C'_A follows x in C_A.

Proof. Lemma 26.6 gives c with $D_B^{(c)}$ refining C_B and $\varepsilon_{B,c} \le \delta$. Both $D_B^{(c)}$ and C_B are chains from P_B to Q_B, so Lemma 26.3 gives a pattern x with $D_B^{(c)}$ following x in C_B. Since Δ_A can follow C_A

above ℓ_A, Definition 26.4 applied to x gives ℓ'_A and C'_A. □

Definition 26.8 (rung). A *rung* for (Δ_1, Δ_2) consists of: a number N; chains C_1 in X_1 and C_2 in X_2, each with N links; levels ℓ_1, ℓ_2 such that C_n is a chain from P_n to Q_n and a consolidation of $D_n^{(\ell_n)}$ $(n = 1, 2)$; and two positive reals mesh and mfollow such that every link of C_1 and of C_2 has diameter $\leq$ mesh, and *at least one* side n satisfies both: Δ_n can follow C_n above ℓ_n, and every link of C_n has diameter $\leq$ mfollow.

The two bounds play different roles: mesh is what Theorem 24.2 will see as ε_i; mfollow is the sharper bound carried by the side that is currently a level, and it becomes the mesh of the *next* rung (Proposition 26.10).

Proposition 26.9 (the base rung). There is a rung with mfollow $= 1/2$.

Proof. Since $\varepsilon_{1,a} \to 0$ and $\operatorname{dist}(P_1, Q_1) > 0$, choose a with $\varepsilon_{1,a} < 1/2$ and $\varepsilon_{1,a} < \operatorname{dist}(P_1, Q_1)$. Then $D_1^{(a)}$ has at least 2 links (a chain from P_1 to Q_1 with one link would have a link of diameter $\geq \operatorname{dist}(P_1, Q_1)$). Since $n_{2,b} \to \infty$ as $b \to \infty$ (Lemma 18.2, using $P_2 \neq Q_2$), choose b with $n_{2,b} \geq n_{1,a}$, and let F be a consolidation of $D_2^{(b)}$ from P_2 to Q_2 with exactly $n_{1,a}$ links: the block map sending index t to $\min(t, n_{1,a} - 1)$ (singleton blocks followed by one long tail block) does this, since P_2 lies in link 0 alone and Q_2 in the last link alone. Take $N = n_{1,a}$, $C_1 = D_1^{(a)}$, $C_2 = F$, $\ell_1 = a$, $\ell_2 = b$, mfollow $= 1/2$, mesh $= \max(1/2, \operatorname{diam} X_2 + 1)$; side 1 can follow $C_1 = D_1^{(a)}$ above a by pattern following, and its links have diameter $\leq \varepsilon_{1,a} < 1/2$. □

Note. The base rung's side 2 is an arbitrary consolidation of the right length; its mesh bound is the crude $\operatorname{diam} X_2 + 1$. This is harmless because the first rung's mesh is not what controls the construction: Theorem 24.2 needs only $\sum_i \varepsilon_i < \infty$, and one large term does not matter.

Proposition 26.10 (the next rung). Let r be a rung and $\delta > 0$. There is a rung r' with mfollow$'$ $= \delta$ and mesh$'$ $= \max(\delta, \text{mfollow})$, together with a pattern $x : \{0, \dots, N' - 1\} \to \{0, \dots, N - 1\}$ such that C'_1 follows x in C_1 and C'_2 follows x in C_2.

Proof. Suppose side 1 is the side of r that can follow (the other case is the mirror image). Apply Proposition 26.7 with $A = 1$, $B = 2$: it gives $c \geq \ell_2$ with $\varepsilon_{2,c} \leq \delta$, a level $\ell'_1 > \ell_1$, a chain C'_1 from P_1 to Q_1 with $n_{2,c}$ links, a consolidation of $D_1^{(\ell'_1)}$, and a pattern x with $D_2^{(c)}$ following x in C_2 and C'_1 following x in C_1. Set $N' = n_{2,c}$, $C'_2 = D_2^{(c)}$, $\ell'_2 = c$. Then: C'_2 is a level, so Δ_2 can follow it above c (pattern following), and its links have diameter $\leq \varepsilon_{2,c} \leq \delta =$ mfollow$'$; the links of C'_1 have diameter $\leq$ mfollow by Lemma 26.2, since C'_1 follows x in C_1 and the links of C_1 (the following side of r) have diameter $\leq$ mfollow. So both sides have mesh $\leq \max(\delta, \text{mfollow}) =$ mesh$'$. □

Note. The side that could follow loses that property in r' (it is now a consolidation of a level, not a level) but keeps its sharp mesh bound; the other side becomes a level, gains the property, and has the new sharp bound δ. That is the alternation of Bing's proof — his $D_1, E_1, D_2, E_2, \dots$ on pp. 738–739 — and the reason the two bounds have to be tracked separately.

27. Proof of Theorem 24.3

The sequence of rungs. Let r_0 be the base rung (Proposition 26.9) and, for $k \geq 0$, let r_{k+1} be the next rung after r_k with $\delta = (1/2)^{k+2}$ (Proposition 26.10), with the pattern x_k joining r_{k+1} to r_k.

Write N_k, $C_{1,k}$, $C_{2,k}$, $\ell_{n,k}$, mesh_k, $\mathrm{mfollow}_k$ for the data of r_k.

The mesh bookkeeping. By construction $\mathrm{mfollow}_k = (1/2)^{k+1}$ for all k (base: $1/2$; step: $\delta = (1/2)^{k+2}$). Hence for $k \geq 0$,

$$\mathrm{mesh}_{k+1} = \max\big((1/2)^{k+2}, \mathrm{mfollow}_k\big) = \max\big((1/2)^{k+2}, (1/2)^{k+1}\big) = (1/2)^{k+1}.$$

So $\sum_k 2\,\mathrm{mesh}_k = 2\,\mathrm{mesh}_0 + \sum_{k\geq 0}(1/2)^k < \infty$.

The matching covers. Put $\varepsilon_i = 2\,\mathrm{mesh}_i$ (positive and summable) and, at level i, N_i indices with $x^1_{i,k}$ the link k of $C_{1,i}$ and $x^2_{i,k}$ the link k of $C_{2,i}$. Conditions of Definition 24.1:

1. (cover) $M_n \subseteq C^*_{n,i}$, since $C_{n,i}$ is a consolidation of a level (Lemma 26.5).
2. (meet) every link of $C_{n,i}$ meets M_n (Lemma 26.5).
3. (diameter) links are bounded (X_n is compact) with diameter $\leq \mathrm{mesh}_i < \varepsilon_i$.
4. (match) $C_{1,i+1}$ follows x_i in $C_{1,i}$ and $C_{2,i+1}$ follows the same x_i in $C_{2,i}$; the links of $C_{2,i}$ have diameter $\leq \mathrm{mesh}_i < \varepsilon_i$; so Lemma 26.1 gives (4) in the direction $1 \to 2$, and the same lemma with the sides exchanged gives the other direction.

M_1 and M_2 are compact (Lemma 3.2). Theorem 24.2 gives a homeomorphism $T : M_1 \to M_2$ with the tracking property.

The end points. The constant index sequence $a_i = 0$ tracks P_1 on side 1 and P_2 on side 2, since $C_{n,i}$ is a chain from P_n to Q_n and so P_n lies in its link 0; hence $T(P_1) = P_2$. Likewise $a_i = N_i - 1$ tracks Q_1 and Q_2, so $T(Q_1) = Q_2$. □

Note (*what the proof uses*). Of the descriptions, only: that each level is a chain from P_n to Q_n; that levels refine earlier levels; that $\varepsilon_{n,i} \to 0$; compactness of X_n (through Lemma 3.2 and Theorem 24.2); and pattern following (Theorem 20.1 for every tail), which is where crookedness enters, once, through Part V. Bing's Theorem 12 is thus: Theorem 6 plus Theorem 11 plus the alternation. The statement that the repository first proved was Theorem 24.3 with pattern following as a hypothesis (`Description.exists_homeomorph`); the unconditional `theorem12` is that statement discharged by Part V.

Part VII. Theorem 13: the described continuum is homogeneous

28. Statement and plan

Throughout this Part, Δ is a description from P to Q in a compact metric space X (Definition 3.1), with $P \neq Q$, chains $D^{(i)}$ with n_i links, constants ε_i, and continuum M.

Definition 28.1 (composant). For $p \in M$, the *composant* of p in M is the union of all proper subcontinua of M containing p: all continua $K \subseteq M$ (Definition 17.1) with $K \neq M$ and $p \in K$.

Theorem 28.2 (Bing, Theorem 13). M is homogeneous: for any $P_1, P_2 \in M$ there is a homeomorphism $T : M \to M$ with $T(P_1) = P_2$.

Bing's proof (p. 740) has three moves. Given $P_1 \in M$, choose $Q_1 \in M$ not in the composant of P_1; this needs the composant to be a proper subset of M, which Bing takes from the indecomposability of M (Theorem 10) and the standard theory of indecomposable continua. Then construct a *description of M from P_1 to Q_1*: a sequence $E_1, E_2, \ldots$ of chains from P_1 to Q_1 satisfying Definition

3.1 with $\bigcap E_i^* = M$. Bing obtains E_{i+1} by taking, at a deep level k, the subchain $D_k(P_1, Q_1)$ of D_k from the link containing P_1 to the link containing Q_1, observing that it meets both end links of the previous level (his first paragraph), and applying Theorem 4 to consolidate it into a chain whose links lie in adjacent links of the previous level. Do the same for P_2, Q_2. Then Theorem 12 applied to the two descriptions gives a homeomorphism of M onto M carrying P_1 to P_2.

The proof below follows this plan. Every step is carried out, and the following points, which Bing's text leaves to the reader, are proved in full: that M is connected at all (§29; Theorem 10 as stated in Part IV is about subcontinua of M and does not need it, but Theorem 13 does); that composants of an indecomposable continuum are proper (§30, the Baire argument); the limit argument of Bing's first paragraph, which needs that a limit of finely chained sets is connected (§29 and §31); and that the two-link basepoint version of Theorem 4 (Theorem 10.2) is what the construction of E_{i+1} requires (§32).

29. Limits of chained sets; M is a continuum

Definition 29.1. A set $A \subseteq X$ is *ε-chained* if whenever $A \subseteq U \cup V$ with every point of U more than ε from every point of V, then $A \subseteq U$ or $A \subseteq V$. For an infinite set $I \subseteq \mathbb{N}$ and sets A_k ($k \in I$), the *limit superior* is

$$\limsup_{k \in I} A_k = \bigcap_N \overline{\bigcup_{k \in I,\, k \geq N} A_k}.$$

It is closed. If $R \in A_k$ for every $k \in I$ then $R \in \limsup A_k$.

Lemma 29.2 (spans are chained). If every link of a chain D is bounded with diameter $\leq \varepsilon$, then for $a \leq b$ the union $D_a \cup \cdots \cup D_b$ of the links of the subchain $D(a, b)$ is ε-chained.

Proof. Let the union lie in $U \cup V$ with U, V more than ε apart. Each link, having diameter $\leq \varepsilon$, lies entirely in U or entirely in V (a link with a point in each would have two points more than ε apart). Adjacent links share a point, so they lie on the same side; by induction along $a, a+1, \ldots, b$ all links lie on the side of D_a. □

Proposition 29.3 (a limit of finely chained sets is connected). Let $\varepsilon_k \to 0$, I infinite, A_k ε_k-chained for every $k \in I$, and $R \in A_k$ for every $k \in I$. Then $Z = \limsup_{k \in I} A_k$ is connected.

Proof. Z is closed and contains R. Let $Z \subseteq u \cup v$ with u, v closed and disjoint; say $R \in u$; we show $Z \subseteq u$. Since u is compact and v is closed and disjoint from it, there is $\delta > 0$ with $\operatorname{dist}(u, v) \geq \delta$. Let U', V' be the open $\delta/2$-neighbourhoods of u, v; every point of U' is at least $\delta/2$ from every point of V' (if $x \in U'$, $y \in V'$ were closer, a point of u and a point of v would be within δ). The closed sets $C_N = \overline{\bigcup_{k \in I, k \geq N} A_k}$ decrease with N, and their intersection Z lies in the open set $U' \cup V'$; by compactness, some $C_{N_1} \subseteq U' \cup V'$. Take N_2 with $\varepsilon_k < \delta/2$ for $k \geq N_2$. For $k \in I$ with $k \geq \max(N_1, N_2)$: $A_k \subseteq U' \cup V'$ and A_k is ε_k-chained with $\varepsilon_k < \delta/2$, so $A_k \subseteq U'$ or $A_k \subseteq V'$; and $R \in A_k \cap u \subseteq U'$, so $A_k \subseteq U'$. Hence $C_{\max(N_1, N_2)} \subseteq \overline{U'}$, which lies in the open δ-neighbourhood of u and so misses v. Therefore $Z \cap v = \emptyset$ and $Z \subseteq u$. □

Proposition 29.4 (M is connected). M is a continuum, and every $D^{(i)*}$ is ε_i-chained.

Proof. $(D^{(i)})^*$ is the union of the whole chain $D^{(i)} = D^{(i)}(0, n_i - 1)$, hence ε_i-chained by Lemma 29.2. By Lemma 3.2, $M = \bigcap_i \overline{(D^{(i)})^*}$, and since the unions decrease ($(D^{(k)})^* \subseteq (D^{(N)})^*$ for $k \geq N$), $\bigcup_{k \geq N} (D^{(k)})^* = (D^{(N)})^*$; so $M = \limsup_{k \in \mathbb{N}} (D^{(k)})^*$. P lies in every $(D^{(k)})^*$. Proposition 29.3 with $I = \mathbb{N}$ gives connectedness; compactness is Lemma 3.2. □

Corollary 29.5. M is an indecomposable continuum (Theorem 17.2 with $K = M$).

Note. Neither Bing's paper nor the repository before batch 81 contained a proof that M is connected. Bing's description paragraph says "$M = \bigcap D_i^*$ is a continuum" without argument; it is the standard fact that a nested intersection of closed connected sets in a compact space is connected, except that the D_i^* are open, so one passes to closures — which are not obviously connected either — or argues as above.

30. Composants are proper

Fix an indecomposable continuum M in the compact space X (in this section, an arbitrary one).

Lemma 30.1 (boundary bumping, clopen form). Let M be connected, $L \subseteq M$ with $L \neq M$, and $W \subseteq L$ nonempty, closed, and relatively open in L ($W = L \cap O$ for an open O). Then W meets $\overline{M \setminus L}$.

Proof. Suppose not. Let $O' = O \setminus \overline{M \setminus L}$, open. Then $W \subseteq O'$ (as $W \subseteq O$ misses the closure), and $M \cap O' \subseteq W$: a point of M in O not in $\overline{M \setminus L}$ lies in $L \cap O = W$. So $W = M \cap O'$ is relatively open in M, and it is closed; M connected and $W \neq \emptyset$ give $W = M$. But $W \subseteq L \neq M$ with $L \subseteq M$. □

Lemma 30.2 (a clopen piece plus a continuum). Let K be closed and connected, L a set such that every nonempty closed relatively open subset of L meets K, and $L_1 \subseteq L$ closed and relatively open in L. Then $K \cup L_1$ is connected.

Proof. Let $K \cup L_1 \subseteq u \cup v$, u, v closed and disjoint. K is connected, so $K \subseteq u$, say (if K is empty the argument below still applies). $L_1 \cap v = L_1 \setminus u$ is closed and relatively open in L (it is $L \cap (O_1 \setminus u)$), so if nonempty it meets $K \subseteq u$, contradicting disjointness. Hence $L_1 \subseteq u$ and $K \cup L_1 \subseteq u$. □

Proposition 30.3 (proper subcontinua have empty interior). Let $K \subseteq M$ be a continuum with $K \neq M$. There is no open set O with $O \cap M \neq \emptyset$ and $O \cap M \subseteq K$.

Proof. Suppose there is. Put $L = \overline{M \setminus K}$, a closed subset of M. Since $O \cap M \subseteq K$, O misses $M \setminus K$, and as O is open it misses L; so $L \neq M$ (L misses the nonempty $O \cap M$). Also $K \cup L = M$ and $\overline{M \setminus L} \subseteq K$ ($M \setminus L \subseteq K$ and K is closed). By Lemma 30.1, every nonempty closed relatively open subset of L meets $\overline{M \setminus L} \subseteq K$.

If L is connected, then $M = K \cup L$ with K, L continua, neither equal to M — contradicting indecomposability. If L is not connected, write $L = (L \cap t) \cup (L \cap t')$ with t, t' closed, $L \cap t \cap t' = \emptyset$, both parts nonempty; then $L \cap t = L \setminus t'$ is relatively open in L, and likewise $L \cap t'$. By Lemma 30.2, $K \cup (L \cap t)$ and $K \cup (L \cap t')$ are connected; they are closed, hence continua; their union is M. Neither is M: a point $z \in L \cap t'$ lies in $\overline{M \setminus K}$ and in the open set $X \setminus t$, so some $y \in M \setminus K$ has $y \notin t$, and $y \notin K \cup (L \cap t)$; symmetrically for the other. Again indecomposability is contradicted. □

Proposition 30.4 (a point outside every proper subcontinuum through p). For every $p \in M$ there is $y \in M$ lying in no proper subcontinuum of M containing p.

Proof. Suppose every $y \in M$ lies in some proper subcontinuum through p. Let $\mathcal{B}$ be a countable base of X and $\mathcal{B}' = \{b \in \mathcal{B} : b \cap M \neq \emptyset\}$. For $b \in \mathcal{B}'$ let C_b be the connected component of p in $M \setminus b$ (empty if $p \in b$). Each C_b is closed (a component of the closed set $M \setminus b$), connected,

contained in $M \setminus b$ and so a proper subset of M. The C_b cover M: given $y \in M$ and a proper subcontinuum $K \ni p, y$, pick $x \in M \setminus K$ and a basic $b \ni x$ with $b \cap K = \emptyset$; then $b \in \mathcal{B}'$, K is a connected subset of $M \setminus b$ containing p, so $K \subseteq C_b$ and $y \in C_b$. Thus the compact space M is a countable union of closed sets C_b; by Baire, some C_b has nonempty interior in M: an open O with $O \cap M \neq \emptyset$ and $O \cap M \subseteq C_b$. C_b is then nonempty, closed in a compact space, connected — a proper subcontinuum with nonempty interior in M, contradicting Proposition 30.3. □

Corollary 30.5 (composants are proper). For every $p \in M$, the composant of p is not all of M; so there is $q \in M$ outside the composant of p.

Lemma 30.6. In the described continuum M (with $P \neq Q$): every $R \in M$ lies in its own composant (the singleton $\{R\}$ is a proper subcontinuum, as M has two points); hence if S is not in the composant of R then $S \neq R$.

31. Subchains reaching both ends

Definition 31.1 (subchain from R to S). For a chain D and indices $a \leq b$, say $D(a, b)$ is a *subchain from R to S* if $D(a, b)$ is a chain from R to S or a chain from S to R (Definition 1.2): R lies in D_a and in no other link of $D(a, b)$, S in D_b and in no other, or the same with R, S exchanged. This is Bing's $D(R, S)$.

Lemma 31.2. Let $R, S \in D^*$ lie in no common link of D. Then some $D(a, b)$ is a subchain from R to S.

Proof. The links containing R form a set I_R of at most two adjacent indices, and likewise I_S; the two sets are disjoint. If some element of I_R were less than some element of I_S and some greater, then between two adjacent elements of I_R there would be no room for an element of I_S, so the sets would interleave impossibly — precisely: all of I_R is below all of I_S or all above. Say below. Let $a = \max I_R$, $b = \min I_S$. In $D(a, b)$, R lies in D_a and in no D_t with $a < t \leq b$ (such $t \notin I_R$); S lies in D_b and in no D_t with $a \leq t < b$. So $D(a, b)$ is a chain from R to S. The other case gives a chain from S to R. □

Lemma 31.3. If $R, S \in M$ and $\varepsilon_k < \operatorname{dist}(R, S)$, then $D^{(k)}$ has a subchain from R to S.

Proof. $R, S \in (D^{(k)})^*$ (Lemma 3.2), and no link of diameter $\leq \varepsilon_k$ contains both; Lemma 31.2. □

Lemma 31.4 (the limit step). Let S be outside the composant of R in M. Let I be infinite and, for $k \in I$, let $A_k \subseteq (D^{(k)})^*$ be ε_k-chained with $R, S \in A_k$. Then there is no open set O meeting M with $A_k \cap O = \emptyset$ for all $k \in I$.

Proof. Suppose there is. Let $Z = \limsup_{k \in I} A_k$. Z is closed, connected (Proposition 29.3) and contains R and S. $Z \subseteq M$: for every i, $Z \subseteq \overline{\bigcup_{k \in I, k \geq i+1} A_k} \subseteq \overline{(D^{(i+1)})^*} \subseteq (D^{(i)})^*$ (levels refine earlier levels; Definition 3.1(3)). $Z \neq M$: a point $w \in O \cap M$, if in Z, would be a limit of points of $\bigcup_k A_k$, and O is an open neighbourhood of w missing every A_k. So Z is a proper subcontinuum of M containing R and S, and S is in the composant of R — contradiction. □

Proposition 31.5 (reaching both ends; Bing's first paragraph). Let S be outside the composant of R in M. For every j there is $k > j$ and a subchain $D^{(k)}(a, b)$ from R to S such that some link of $D^{(k)}(a, b)$ meets $D_0^{(j)}$ and some link meets $D_{n_j - 1}^{(j)}$.

Proof. $R \neq S$ (Lemma 30.6), so $\varepsilon_k < \operatorname{dist}(R, S)$ for $k \geq k_0$. For each $k \geq \max(k_0, j + 1)$ fix a subchain $D^{(k)}(a_k, b_k)$ from R to S (Lemma 31.3) and let A_k be the union of its links: ε_k-chained

(Lemma 29.2), contained in $(D^{(k)})^*$, containing R and S. Suppose the conclusion fails for every such k: then for each k, A_k misses $O_1 = D_0^{(j)}$ or misses $O_2 = D_{n_j-1}^{(j)}$. One of the two index sets $I_1 = \{k : A_k \cap O_1 = \emptyset\}$, $I_2 = \{k : A_k \cap O_2 = \emptyset\}$ is infinite. Each O_i is open and meets M (Lemma 3.2). Lemma 31.4 applied to that I_i and O_i gives a contradiction. □

Note. Bing writes: "there is an integer k such that $D_k(P_1, Q_1)$ intersects both the first and the last link of D_j; for otherwise the limiting set of the $D_k(P_1, Q_1)$ would be a proper subcontinuum of M containing P_1 and Q_1." The two things the paragraph assumes are that the limiting set is connected (Proposition 29.3, whose hypothesis is the chainedness of the spans, Lemma 29.2) and that it is proper because it misses an open set that meets M (the last sentence of Lemma 31.4). The infinite subsequence on which the same end link is missed is where "the limiting set" is made precise.

32. The description from R to S

Proposition 32.1 (Theorem 4 on a subchain). Let E be crooked in D, where D has $k \geq 5$ links, and let $E(a, b)$ be a chain from R to S such that some link of $E(a, b)$ meets D_0 and some link meets D_{k-1}. Then there is a block map β on *all* of E such that E/β is a chain from R to S (Definition 1.2) and every link of E/β lies in the union of three consecutive links of D.

Proof. The links of E containing R are among E_{a-1}, E_a ($R \in E_a$, and R is in no other link of $E(a, b)$, so any other link containing R is adjacent to E_a and outside the subchain, hence E_{a-1}); likewise the links containing S are among E_b, E_{b+1}. Put $r = \max(a - 1, 0)$, $r' = r + 1$, $s = \min(b + 1, n - 1)$, $s' = s - 1$. The links meeting D_0 and D_{k-1} are not adjacent (their images in D are at least $k - 3 \geq 2$ apart), so $b \geq a + 2$ and $r' \leq s'$. Theorem 10.2 applied to r, r', s', s (the meeting links lie in $[a, b] \subseteq [r, s]$) gives β with $\beta(r) = \beta(r') = 0$, $\beta(s') = \beta(s) = \max \beta$, and three-link blocks. Every link containing R has index r or r' (if $a = 0$ then $R \in E_0$ and possibly E_1; if $a \geq 1$, indices $a - 1 = r$ and $a = r'$), so has $\beta = 0$; every link containing S has $\beta = \max \beta$. By Lemma 16.1, E/β is a chain from R to S. □

This is the point at which the two-link basepoint version of Theorem 4 (Theorem 10.2, Corollary 16.2) is needed: R may lie in two links of E, only one of which is in Bing's subchain $E(R, S)$, and both must go into the first block.

Proposition 32.2 (one step of Bing's sequence). Let S be outside the composant of R. For every level L there are $k > L$ and a chain F from R to S which is a consolidation of $D^{(k)}$, such that the closure of every link of F lies in some link of $D^{(L)}$ and every link of F has diameter $\leq 1/(L + 1)$.

Proof. The links of $D^{(L)}$ cover the compact set $\overline{(D^{(L+1)})^*}$ (Definition 3.1(3)); let δ be a Lebesgue number: every subset of that compact set of diameter $< \delta$ lies in a single link of $D^{(L)}$. Put $\eta = \min(\delta, 1/(L+1))$. Choose $m \geq L+1$ with $\varepsilon_m < \eta/3$ and $n_m \geq 5$ (the link counts tend to infinity as $P \neq Q$, Lemma 18.2). Proposition 31.5 with $j = m$ gives $k > m$ and a subchain $D^{(k)}(a, b)$ from R to S meeting $D_0^{(m)}$ and $D_{n_m-1}^{(m)}$. $D^{(k)}$ is crooked in $D^{(m)}$ (Lemma 3.4). If $D^{(k)}(a, b)$ is a chain from R to S, Proposition 32.1 gives $F = D^{(k)}/\beta$, a chain from R to S, a consolidation of $D^{(k)}$ (Lemma 1.5), each link inside three consecutive links of $D^{(m)}$; if it is a chain from S to R, apply Proposition 32.1 with R, S exchanged and reverse the resulting chain. Each link of F then has diameter $\leq 3\varepsilon_m < \eta \leq 1/(L + 1)$ (Lemma 18.2). Its closure lies in the closed $3\varepsilon_m$-ball about any of its points x_0; $x_0 \in (D^{(k)})^* \subseteq (D^{(L+1)})^*$, and the ball has diameter $< \delta$, so it lies in a link of $D^{(L)}$. □

Proposition 32.3 (a description from R to S). Let S be outside the composant of R. There is a description Δ' from R to S in X with the same continuum: $M' = M$.

Proof. For each L choose $k(L) > L$ and $F(L)$ as in Proposition 32.2. Define levels $\ell_0 = 0$, $\ell_{i+1} = k(\ell_i) + 1$; then $\ell_i \geq i$. Let Δ' have chains $D'^{(i)} = F(\ell_i)$ and constants $\varepsilon'_i = 1/(i+1)$. Check Definition 3.1:

1. $F(\ell_i)$ is a chain from R to S.
2. Crookedness: $F(\ell_{i+1})$ has the closure of each link in a link of $D^{(\ell_{i+1})} = D^{(k(\ell_i)+1)}$, so refines it; $D^{(k(\ell_i)+1)}$ is crooked in $D^{(k(\ell_i))}$; and $F(\ell_i)$ is a consolidation of $D^{(k(\ell_i))}$. By Theorem 2.3 (crookedness passes to a consolidation of the coarse chain), $D^{(k(\ell_i)+1)}$ is crooked in $F(\ell_i)$; by Theorem 2.4, so is its refinement $F(\ell_{i+1})$.
3. Closures: the closure of a link of $F(\ell_{i+1})$ lies in a link of $D^{(k(\ell_i)+1)}$, whose closure lies in a link of $D^{(k(\ell_i))}$, which lies in a link of $F(\ell_i)$.
4. Diameters: $\leq 1/(\ell_i+1) \leq 1/(i+1) = \varepsilon'_i$; links are bounded, being inside links of $D^{(\ell_i)}$; $\varepsilon'_i \to 0$.

The continuum: $F(\ell_i)^* = (D^{(k(\ell_i))})^*$ (a consolidation has the same union). So $\bigcap_i F(\ell_i)^* = \bigcap_i (D^{(k(\ell_i))})^*$, which equals $M = \bigcap_j (D^{(j)})^*$ because the $k(\ell_i)$ are unbounded and the unions decrease. □

Note. Level 0 of Δ' is $F(0)$, a consolidation of $D^{(k(0))}$ with $k(0) > 0$ — not $D^{(0)}$ itself. Bing's "$E_1 = D_1$" is not available: $D^{(0)}$ is a chain from P to Q, not from R to S.

Proof of Theorem 28.2. Let $P_1, P_2 \in M$. M is an indecomposable continuum (Corollary 29.5). By Corollary 30.5 choose $Q_1 \in M$ outside the composant of P_1 and $Q_2 \in M$ outside the composant of P_2; then $P_n \neq Q_n$ (Lemma 30.6). Proposition 32.3 gives descriptions Δ_1 from P_1 to Q_1 and Δ_2 from P_2 to Q_2, both with continuum M. Theorem 24.3 gives a homeomorphism $T : M \to M$ with $T(P_1) = P_2$ (and $T(Q_1) = Q_2$). □

Note (*what the proof uses*). Of Bing's paper: Theorems 1 and 2 (Part I) in the construction of Δ'; Theorem 4 in its two-link form (Part III) at each step; Theorem 10 (Part IV) for indecomposability; Theorem 12 (Part VI, hence Parts II and V) at the end. Compactness of X enters through Lemma 3.2, the Lebesgue number of Proposition 32.2, and Propositions 29.3, 30.4 and Theorem 24.2. Nothing about the plane is used anywhere.

33. Summary of the dependencies

Theorem 28.2 (Bing 13)	$\Leftarrow$	24.3 (Bing 12), 32.3, 30.5, 29.5
Theorem 24.3 (Bing 12)	$\Leftarrow$	24.2 (Bing 11), 20.1 (Bing 6, every tail)
Theorem 20.1 (Bing 6)	$\Leftarrow$	21.5 (combinatorial core), 3.5 (thick readings)
Proposition 32.3	$\Leftarrow$	32.2 $\Leftarrow$ 31.5, 10.2 (Bing 4, two-link form), 2.3, 2.4
Theorem 10.2 (Bing 4)	$\Leftarrow$	Part II (Bing 3: the fold), §§12–15
Corollary 29.5	$\Leftarrow$	17.2 (Bing 10), 29.4
Corollary 30.5	$\Leftarrow$	30.3, Baire
Proposition 31.5	$\Leftarrow$	29.3, 31.4

Theorem 20.3 (Bing's Theorem 5) and Bing's Theorems 7–9 are not on any path to Theorem 28.2.

Appendix A. Correspondence with the Lean development

All names are in namespace `HPC`; files are under `HPC/`. Part II's items are listed with their own prefix `II.`; the conventions that differ between Part II and its Lean are recorded at the end of that Part's table.

Part I

This document	Lean
Def. 1.1	`Chain` (`Chains/Basic.lean`): fields `link`, `isOpen`, `adjacent`; `Chain.union`, `Chain.sub`, `Chain.reverse` (`Chains/Consolidation.lean`)
Def. 1.2	`Chain.IsFrom`
Def. 1.3	`Chain.Refines`, `Chain.IsConsolidationOf`
Def. 1.4, Lemma 1.5	`IsBlockMap`, `Chain.ofBlocks`, `Chain.ofBlocks_isConsolidationOf`, `Chain.blockLink_adjacent` (`Chains/Consolidation.lean`)
Def. 1.6	`Chain.FollowsPattern`; `Description.IsPattern` (`PseudoArc/Description.lean`)
Lemma 1.7	`Chain.pattern_lipschitz`
Def. 2.1, 2.2	`Chain.Between`, `Chain.IsCrookedIn'` (the primed form is the one used throughout; `IsCrookedIn'.isCrookedIn` relates it to the literal `IsCrookedIn`)
Thm. 2.3	`Chain.isCrookedIn'_of_isConsolidationOf`
Thm. 2.4	`Chain.isCrookedIn'_of_refines`
Cor. 2.5	`Chain.isCrookedIn'_of_isConsolidationOf'`
Lemma 2.6	`IsCrookedIn'.sub`, `IsCrookedIn'.reverse_left`, `IsCrookedIn'.reverse_right`, `IsCrookedIn'.restrict`
Def. 3.1	`Description` (`PseudoArc/Description.lean`): fields `n`, `D`, `isFrom`, `crooked`, `closure_subset`, `ε`, `ε_pos`, `ε_tendsto`, `bdd`, `diam_le`; `Description.M`
Lemma 3.2	`P_mem_M`, `Q_mem_M`, `isClosed_M`, `isCompact_M`, `M_subset_union`, `M_eq_iInter_closure` (`T13Chains.lean`), `link_inter_M_nonempty`, `closure_union_subset`, `refines_of_le`
Def. 3.3	`Description.shift` (`PseudoArc/Statements79.lean`)
Lemma 3.4	`Description.crooked_of_lt`
Lemma 3.5	`Description.eventually_thick` (`PseudoArc/PatGeom.lean`); `eventually_thick_at` (`T5Main.lean`)

Part II

All names are in namespace `HPC.Fold` (files under `HPC/Fold/`) unless noted.

This document	Lean
Def. II.0.1	`HPC.Chain.IsCrookedIn'` (`Chains/Basic.lean`)
Def. II.0.2	`IsBlockMap`, `Chain.ofBlocks` (`Chains/Consolidation.lean`)
Thm. II.0.3	`theorem3` (`Open.lean`)
§II.1, ρ, σ	`Refines.minIdx`, `Refines.maxIdx`
Lemma II.1.1	`IsCrookedIvF` (`Faithful.lean`), fields `le`, `width`, `crooked`; `minIdx_lip`; `isCrookedIvF_minIdx_maxIdx`
Def. II.2.2	`FoldsOnto` (`Itinerary.lean`)
Prop. II.2.3	`exists_blocks_first_of_folds` (`Itinerary.lean`)
Prop. II.2.4	`exists_blocks_first_of_range_le_two`
Thm. II.2.5	`S3` (`Span.lean`); `s3_general` (`General.lean`) `:= Z77.s3_zig`
Thm. II.0.3 from II.2.5	`theorem3_of_S3` (`Span.lean`)
§II.3, S, arms	`Z77.reindex`, `Z77.arm`, `Z77.sside` (`Zig77Crook.lean`, `Zig77Main.lean`)
Def. II.3.1	`Z77.CrkOn`
Lemma II.3.2	`Z77.crkOn_of_isCrookedIvF`
Lemma II.3.3	`Z77.CrkOn.reverse`, `Z77.crkOn_sside`
Lemma II.3.4	`Z77.up_core`
Lemma II.3.5	`Z77.down_core`
Def. II.4.1	`Z77.legLen`, `Z77.jwalk`, `Z77.jlen` (`Zig77Walk.lean`)
Def. II.4.2	`Z77.Env`, `Z77.arr`
Def. II.4.3	`Z77.Fit`
Lemma II.4.4 and leg position	`Z77.fold_of_fit`, `Z77.legPos`, `Z77.legPos_le`, `Z77.legPos_lip`, `Z77.jwalk_legPos`
§II.5 states	`Bool → Option ℕ` (`none` = finished, `some τ` = live at τ)
Lemma II.5.2	`Z77.first_drop` (`Zig77Step.lean`)
Lemma II.5.3	`Z77.transfer`
Prop. II.5.4 and v_e	`Z77.up_step`, `Z77.ArmVal`, `Z77.exists_armVal`, `Z77.max_on`
Prop. II.5.5	`Z77.down_step`
Lemma II.6.1	`Z77.nofail_up` (`Zig77Main.lean`)
Lemma II.6.2	`Z77.nofail_down`
Lemma II.6.3	`Z77.first_direction`
Prop. II.6.4	`Z77.main_up` (strong induction on `((H + 1 - a) + (b - L + 1)).toNat`)
Prop. II.6.5	`Z77.greedy_fit`

This document	Lean
Proof of Thm. II.2.5	`Z77.s3_zig`, `Z77.foldsOnto_of_nat`

Conventions that differ between the text and the Lean: the Lean indexes arms by `Bool` (`true` = forward, `false` = backward) and reads the "left arm" of §II.3 as `fun x => S (-x)` through `sside`; `Fit` is stated with the turning levels as a `List ℤ` and with the arm read on $\mathbb{N}$, cut off at `m`; `FoldsOnto` is stated for `Fin m → ℤ`, and `foldsOnto_of_nat` transfers the $\mathbb{N}$-indexed fold. The hypotheses $\exists t, \rho(t) = L$ and $\exists t, \rho(t) = H$ carried by `S3` are not used by `s3_zig`. The "left arm" of §II.3 is read as `fun x => S (-x)` through `sside`.

Part III

Names in namespace `Fold` (files `Fold/`).

This document	Lean
Thm. 10.1	`theorem4` (`Theorem4.lean`)
Thm. 10.2	`theorem4_pair` (`Theorem4Pair.lean`); `B79.theorem4_pair_core` (`Pair79Assemble.lean`)
Lemma 12.1	`B78.meet_low_before` (`Splice78.lean`)
Def. 12.2, Prop. 12.4	`B78.fold_low_end`
Cor. 12.5	`B78.prefix_fold`, `B78.suffix_fold`
Prop. 13.1	`B78.splice_schedule`
Lemma 13.2	`B78.assemble`, `B78.link_sub_of_level`
Prop. 13.3	`B78.theorem4_core`
Lemma 14.1	`B78.two_window`, `B78.unionRange_reverse`
Lemma 15.1, Def. 15.2	`B79.fold_low_end_mode`, `B79.fold_high_end_mode` (`Pair79Ends.lean`); `B79.EvUp`, `B79.EvDown` (`Pair79Seq.lean`)
Prop. 15.3	`B79.pair_exclusion` (`Pair79Excl.lean`)
Prop. 15.4	`B79.pair_prefix_fold`, `B79.pair_suffix_fold`, `B79.splice_schedule_pair`, `B79.assemble_pair`, `B79.pair_windows`
Lemma 16.1, Cor. 16.2	`Q78.isFrom_ofBlocks_iff`, `Q78.isFrom_ofBlocks_of_theorem4` (`Questions78.lean`)

Part IV

This document	Lean
Def. 17.1	`Fold.Q78.IsContinuum`, `Fold.Q78.IsIndecomposable` (`Fold/Questions78.lean`); `Fold.Q78.Theorem10Statement`
Thm. 17.2	`Description.theorem10`, `Description.theorem10_core` (`PseudoArc/Theorem10.lean`)
Lemma 18.1	`Chain.inter_link_nonempty_between`
Lemma 18.2	`Chain.dist_le_of_mem_link`; `Chain.dist_le_of_mem_unionRange`, `Chain.dist_le_two` (`T13Step.lean`); `Description.n_tendsto_atTop`, `two_le_n_of_lt`
Cor. 19.1	`Description.isContinuum_M` (`T13Chains.lean`) with `theorem10`

Part V

Files `PseudoArc/Pat*.lean`, `T5*.lean`, `Theorem5.lean`, `Statements79.lean`.

This document	Lean
Thm. 20.1	`Description.theorem6` (`Theorem6Statement`), `theorem6Pos_shift` (`Theorem6PosStatement`), `Description.patternFollowing` (`Description.PatternFollowing`)
Prop. 20.2	`not_theorem6Pos_shift_unconditional`, `not_patternFollowing_unconditional`, `unitDescription_not_theorem6`
Thm. 20.3	`Description.theorem5` (`Theorem5Statement`), `theorem5_of` (`T5Main.lean`)
Defs. 21.1–21.4	`PatCore.Al`, `PatCore.Real`, `PatCore.Setting`, `PatCore.Slow`, `PatCore.Good` (`PatCore.lean`); `PatCore.IH`
Thm. 21.5	`PatCore.realize`, `PatCore.ih_all` (`PatMain.lean`)
Cor. 21.6	`PatCore.run_lift`
Lemma 22.1, Prop. 22.2	`theorem6Pos_of_ne` (`PatGeom.lean`)
§23	`T5.rd` (`T5Read.lean`), `T5Runs.lean`, `T5.assemble` (`T5Chain.lean`), `T5Assemble.lean`, `theorem5_of`

Part VI

This document	Lean
Def. 24.1	`MatchingCovers` (`Covers/Matching.lean`), fields `cover`, `meet`, `bdd`, `diam`, `match` (each with ₁, ₂); `MatchingCovers.Tracks₁`, `Tracks₂`; `MatchingCovers.swap`
Thm. 24.2	`MatchingCovers.exists_homeomorph`
Thm. 24.3	`theorem12` (`PseudoArc/Theorem5.lean`), via `Description.exists_homeomorph` (`PseudoArc/Uniqueness.lean`)
Def. 25.1, Lemma 25.2	`tail`, `rad`, `mesh`, `fat`; `rad_eq`, `rad_pos`, `mesh_tendsto_zero`, `exists_mesh_lt`, `dist_le_mesh₂`, `eq_of_forall_dist_le_mesh`
Lemma 25.3	`closure_fat_subset_fat`
Lemma 25.4	`MatchingCovers.step1₂`, `step1₁`
Lemma 25.5	`MatchingCovers.step2₂`, `step2₁`
Lemma 25.6	`MatchingCovers.exists_limit_point`
Def. 25.7	`MatchingCovers.trackSeq`, `toFun`, `toMap`
Lemma 25.8	`MatchingCovers.toFun_mem_y₂`
Props. 25.9–25.11	`continuous_toMap`, `injective_toMap`, `surjective_toMap`
Lemma 26.1	`Chain.pattern_matching` (`Uniqueness.lean`)
Lemma 26.2	`Chain.diam_le_of_followsPattern`, `Chain.diam_le_of_refines`
Lemma 26.3	`Chain.exists_pattern_of_refines`
Def. 26.4	`Description.CanFollow`, `canFollow_level`
Lemma 26.5	`M_subset_consolidation`, `consolidation_link_inter_M_nonempty`
Lemma 26.6	`Description.exists_level_refining` (`Description.lean`)
Prop. 26.7	`Description.exists_half_step`
Def. 26.8	`Rung`
Prop. 26.9	`Rung.exists_base`; `Chain.exists_consolidation_isFrom` (`Chains/Basic.lean`)
Prop. 26.10	`Rung.exists_next`
§27	the body of `Description.exists_homeomorph`

Part VII

Files `PseudoArc/T13Topo.lean`, `T13Chains.lean`, `T13Step.lean`, `Theorem13.lean`, `Statements79.lean`.

This document	Lean
Def. 28.1	`composant` (`Statements79.lean`)
Thm. 28.2	`Description.theorem13` (`Theorem13Statement`)

This document	Lean
Def. 29.1	`T13.ChainedAt`, `T13.limSup`, `T13.isClosed_limSup`, `T13.mem_limSup`
Lemma 29.2	`Chain.span`, `Chain.chainedAt_span`
Prop. 29.3	`T13.isPreconnected_limSup`, `T13.limSup_subset_of_mem`
Prop. 29.4	`Description.isConnected_M`, `chainedAt_union`, `P_mem_union`, `M_eq_iInter_closure`
Cor. 29.5	`Description.isContinuum_M` with `theorem10` (in the proof of `theorem13`)
Lemma 30.1	`T13.clopen_meets_closure_diff`
Lemma 30.2	`T13.isPreconnected_union_clopen`
Prop. 30.3	`T13.false_of_interior`
Prop. 30.4	`T13.exists_not_mem_proper_subcontinua`
Cor. 30.5	`composant_ne_of_indecomposable` (`Theorem13.lean`)
Lemma 30.6	`Description.mem_composant_self`, `ne_of_not_mem_composant`
Def. 31.1	`Chain.IsSubchainFrom` (`Statements79.lean`)
Lemma 31.2	`Chain.exists_isSubchainFrom`, `Chain.val_close_of_mem`
Lemma 31.3	`Description.exists_isSubchainFrom_of_lt`
Lemma 31.4	`Description.false_of_disjoint_limit`
Prop. 31.5	`Description.exists_subchain_meets_ends'`, `exists_subchain_meets_ends`
Prop. 32.1	`Chain.exists_blockChain` (`T13Step.lean`)
Prop. 32.2	`Description.exists_step`
Prop. 32.3	`Description.exists_description_from'`, `exists_description_from`
Proof of 28.2	the body of `Description.theorem13`

Two conventions of the Lean differ from the text. Indices are `Fin n` and statements about "the link containing R" are phrased with the index bounds that `Fin` requires (for example the index `a.val - 1` with natural-number subtraction in `exists_blockChain`, which is what Proposition 32.1's $\max(a-1,0)$ records). The hypothesis `hnd : M.Nontrivial` of `composant_ne_of_indecomposable` is not used by its proof and is not stated in Corollary 30.5.

Appendix B. Verification record

The Lean development is the repository `HPC` (Lean 4, Mathlib, toolchain `leanprover/lean4:v4.28.0`), built with `lake build HPC`. The statements of this document were completed across five deliveries, each accompanied by a `#print axioms` audit file produced by `lake env lean AxiomsNN.lean`:

Delivery	Content completed	Audit file	Declarations audited	Build
batch 77	Part II (Theorem 3 at all spans)	`axioms-batch77.txt`	`theorem3`, `s3_general`, `Z77.s3_zig` and the 34 declarations of the `Zig77` files	clean, 8190 jobs
batch 78	Part III, Theorem 10.1	`axioms-batch78.txt`	17, including `theorem4`, `B78.theorem4_core`, `B78.two_window`	clean, 8190 jobs
batch 79	Part III, Theorem 10.2; Part IV	`axioms-batch79.txt`	33, including `theorem4_pair`, `theorem10`, `inter_link_nonempty_between`, `pair_exclusion`	clean, 8203 jobs
batch 80	Part V; Theorem 24.3 unconditional	`axioms-batch80.txt`	13, including `theorem5`, `theorem6`, `patternFollowing`, `theorem12`, `PatCore.realize`, `T5.assemble`, and the two refutations of Prop. 20.2	clean, 8213 jobs
batch 81	Part VII	`axioms-batch81.txt`	8: `theorem13`, `exists_description_from`, `exists_subchain_meets_ends`, `exists_step`, `composant_ne_of_indecomposable`, `isConnected_M`, `T13.isPreconnected_limSup`, `T13.false_of_interior`	clean, 8217 jobs
batch 84	existence of a description in the plane (not expounded here)	`axioms-batch84.txt`	17, including `Plane.exists_description_plane`, `Plane.pseudoArc_exists`, `Plane.theorem4` (Bing 1951, Theorem 4), `Plane.rectStepSkeleton_false`	clean

Delivery	Content completed	Audit file	Declarations audited	Build
batch 88	Bing, Pacific J. Math. 1951, Theorem 1 (not expounded here)	`axioms-batch88.txt`	22, including `Plane.theorem1_HIC`, `Plane.pseudoArc_unique`, `Plane.isPseudoArc_iff_endPoints`, `PA51.eventually_crookedIn`	clean

Every audited declaration reports exactly `[propext, Classical.choice, Quot.sound]`; none reports `sorryAx`, and `native_decide` is not used. Theorems 24.2 (`MatchingCovers.exists_homeomorph`) and 24.3 in its conditional form (`Description.exists_homeomorph`) were completed in earlier deliveries; they are dependencies of `theorem12`, whose audit in batch 80 covers them transitively. Part I's statements are dependencies of everything above and are covered likewise.

The one `sorry` in the repository is `Fold.theorem3_end`, Bing's end-link clause of Theorem 3 (`axioms-batch78.txt` and later report it with `sorryAx`). It is not a dependency of any statement in this document; the proof of Theorem 10.1 was arranged (§12) so as not to need it.

The audits are Aristotle's. Independent reproduction of the build and of the `#print axioms` output — by the author of the exposition, on a clean checkout — has not yet been carried out. Until it has, every "machine-checked" in this document means "reported as compiled and audited in the delivery named above".

Appendix C. The two statements of Bing 1948 that remain open

Everything in Bing's paper other than the following two statements is machine-checked (Appendix B), or refuted and corrected (Theorem 8, Appendix B). This appendix records the two exceptions as they stand after batches 83–87, with the reductions and refutations that are themselves machine-checked.

C.1. Theorem 14, and with it Theorem 15. Bing's Theorem 14 (p. 740): for triples (P_i, Q_i, R_i) in M with P_i, R_i in one composant and Q_i in another, a homeomorphism of M carries P_1, Q_1, R_1 to P_2, Q_2, R_2. Theorem 15 (subcontinua W_i with $P_i \in W_i$ and Q_i outside the composant of W_i) is derived from it. The following are proved (`HPC/PseudoArc/Theorem14.lean`, `T14Irred.lean`, `Recon86*.lean`, `Recon87*.lean`; standard axioms):

- Irreducible subcontinua exist, and the subcontinuum of M irreducible between two given points is unique (Bing's last paragraph, `irreducible_unique`).
- Theorem 15 follows from Theorem 14 (`theorem15_of_theorem14`), with two hypotheses Bing leaves implicit: $W_i \neq M$, and $P_i \neq R_i$ in Theorem 14; each is necessary.
- Bing's first paragraph, "such sequences exist as a consequence of Theorem 9", reduces to one statement at one level, `covering_step`: *for every L there are a deeper level k and a consolidation F of $D^{(k)}$, a chain from P' to Q' with small links nested in $D^{(L)}$, whose initial subchain from P' to R covers the continuum W irreducible between P' and R.* Given `covering_step`, the description Bing needs exists (`exists_description_covering_of_step`). Theorem 9 as stated does not give `covering_step`: its conclusion controls which links come *after* R's block, and W follows folds of $D^{(k)}$ past R's link.
- `covering_step` in turn follows from a *time-ordering fold* with three-link blocks (`coveringStep_of_folds`$_3$): a block map of $D^{(k)}$ with every block in three consecutive links of a deep $D^{(m)}$, P' alone at

block 0, Q' alone at the last block, R's links at block b, and every link meeting W at a block $\leq b$.

- Without a separation hypothesis the time-ordering fold fails: a seven-link crooked level sequence with the window property (which irreducibility does give, `window_of_irreducible`) admits no such fold (`model_covering_fold₃_refuted`). The failure places Q''s link next to W's links, which cannot happen at deep levels ($Q' \notin W$). Under the corresponding separation hypothesis no failure was found in an exhaustive search through five coarse links and in Bing's chains through seven, but the statement is not proved, and its non-degenerate instances begin at six coarse links.
- P' and R are *opposite end points* of W in the sense of Bing 1951, §5: for every ε there is an ε-chain from P' to R covering W, its links unions of links of a deep level (`exists_chain_from_of_irreducible`). Bing 1951 obtains this from an external theorem; here it follows from crookedness and irreducibility alone. It is the part of the time-ordering fold on W; what is missing is the extension past R to Q'.
- A natural intermediate — a fold of the W-part from P''s link as basepoint with *both* arm ends at the maximal time — is false in general (`model_end_time_refuted`, `model_end_time_refuted'`).

Status. Theorem 14 is not established by Bing's proof as written; whether it is true is not decided here. Theorem 15 depends on it.

C.2. The end-link clause of Theorem 3. Bing's Theorem 3 (p. 732) has a fourth conclusion: a link of the consolidation containing an end link of E that meets D_0 lies in the first links of D. In the three-link block reading this is `theorem3_end`. It is proved when the basepoint is an end link, for injective block maps, and for three-link D (`Recon86End.lean`). Exhaustive search over all crooked level sequences with four coarse links through thirteen links and five coarse links through sixteen, and over the subchains of Bing's minimal chains through seven coarse links, found no basepoint at which every three-link fold violates the clause; for four coarse links and at most seven links this is checked in Lean (`Census87Checks.lean`, with `native_decide`, hence with the axioms `Lean.ofReduceBool` and `Lean.trustCompiler` — the only use of `native_decide` in the development outside the batch-72 audit file, and a dependency of nothing in this document). The clause is not used by Theorem 4 (Part III arranged its proof to avoid it) or by anything else in the paper.

Status. Open; consistent with the evidence; not needed.